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# Distinguishing mutants that resist drugs via different mechanisms by examining fitness tradeoffs across hundreds of fluconazole-resistant yeast strains

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## Abstract

There is growing interest in designing multidrug therapies that leverage tradeoffs to combat resistance. Tradeoffs are common in evolution and occur when, for example, resistance to one drug results in sensitivity to another. Major questions remain about the extent to which tradeoffs are reliable, specifically, whether the mutants that provide resistance to a given drug all suffer similar tradeoffs. This question is difficult because the drug-resistant mutants observed in the clinic, and even those evolved in controlled laboratory settings, are often biased towards those that provide large fitness benefits. Thus, the mutations (and mechanisms) that provide drug resistance may be more diverse than current data suggests. Here, we perform evolution experiments utilizing lineage-tracking to capture a fuller spectrum of mutations that give yeast cells a fitness advantage in fluconazole, a common antifungal drug. We then quantify fitness tradeoffs for each of 774 evolved mutants across 12 environments, finding these mutants group into 6 classes with characteristically different tradeoffs. Their unique tradeoffs may imply that each group of mutants affects fitness through different underlying mechanisms. Some of the groupings we find are surprising. For example, we find some mutants that resist single drugs do not resist their combination, and some mutants to the same gene have different tradeoffs than others. These findings, on one hand, demonstrate the difficulty in relying on consistent or intuitive tradeoffs when designing multidrug treatments. On the other hand, by demonstrating that hundreds of adaptive mutations can be reduced to a few groups with characteristic tradeoffs, our findings empower multidrug strategies that leverage tradeoffs to combat resistance. Finally, by grouping mutants that likely affect fitness through similar underlying mechanisms, our work guides efforts to map the phenotypic effects of mutation.

### eLife assessment

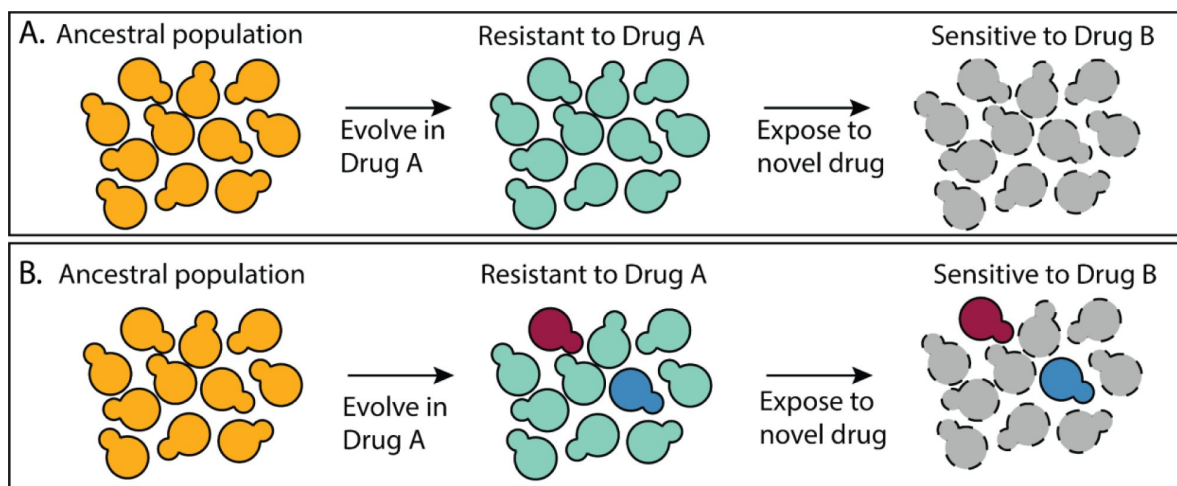
This study provides **valuable** new insights into the trade-offs associated with the evolution of drug resistance. The authors use a **solid** approach to evolve and phenotype hundreds of independent strains. They identify distinct phenotypic clusters based on growth across defined conditions that suggest that tradeoffs are diverse but at the same time could be limited to a few classes based on the underlying resistance mechanisms. The methodologies used align with the current state-of-the-art, and the data and analysis are **solid** as they broadly support the claims, with only minor weaknesses. This work will interest molecular biologists working on the evolution of new phenotypes and microbiologists studying multi-drug therapy.

## Introduction

How many different molecular mechanisms can a microbe exploit to adapt to a challenging environment? Answering this question is particularly urgent in the field of drug resistance because infectious populations are adapting to available drugs faster than new drugs are developed (Ventola 2015 [↗](#); Centers for Disease Control and Prevention (U.S.) 2019 [↗](#)). Understanding the mechanistic basis of drug resistance can inform strategies for how to combine existing drugs in a way that prevents the evolution of resistance (Andersson and Hughes 2010 [↗](#); Melnikov et al. 2020 [↗](#); Pinheiro et al. 2021 [↗](#)). For example, one strategy exposes an infectious population to one drug (Drug A) knowing that the mechanism of resistance to Drug A makes cells susceptible to Drug B (Hall et al. 2009 [↗](#); Pál et al. 2015 [↗](#); Baym et al. 2016 [↗](#); Roemhild et al. 2020 [↗](#)). Problematically, these multi-drug strategies perform best when all mutants that resist Drug A have the same tradeoff in the second drug (**Figure 1A** [↗](#)). If there are multiple different mechanisms to resist Drug A, some of which lack this tradeoff, treatment strategies could fail (**Figure 1B** [↗](#)), and they sometimes do (Grier et al. 2003 [↗](#); Abel zur Wiesch et al. 2014 [↗](#); Wang et al. 2019 [↗](#); Scarborough et al. 2020 [↗](#)).

Laboratory experiments that have power to search for universal tradeoffs – where all the mutants that perform well in one environment perform poorly in another – often find there are mutants that violate trends or the absence of trends altogether (Hill et al. 2015 [↗](#); Nichol et al. 2019 [↗](#); Kinsler et al. 2020 [↗](#); Gjini and Wood 2021 [↗](#); Ardell and Kryazhimskiy 2021 [↗](#); Herren and Baym 2022 [↗](#)). Another way to phrase this observation is to say that adaptive mutations often have collateral effects in environments other than the one in which they originally evolved (Pál et al. 2015 [↗](#)). These effects, also referred to as pleiotropic effects, can be unpredictable and context dependent (Jerison et al. 2020 [↗](#); Geiler-Samerotte et al. 2020 [↗](#); Bakerlee et al. 2021 [↗](#); Chen et al. 2023 [↗](#)). In simpler terms, some mutants that resist Drug A will suffer a tradeoff in Drug B, but others may suffer a tradeoff in Drug C. To sum, observations from many fields suggest that the mutations that provide a benefit in one environment do not always suffer similar tradeoffs. This begs questions about the extent of diversity among adaptive mutations: does each one suffer a unique set of tradeoffs or can many adaptive mutants be grouped by their common tradeoffs? If there are only a few types of tradeoff present in a collection of adaptive mutations, multidrug treatments that target tradeoffs may be more feasible.

The goal of the present study is to count how many different types of adaptive mutation, each type being defined by its unique tradeoffs, exist in a population of drug-resistant yeast. This simple counting exercise is surprisingly difficult. One reason why is that the mutations that provide the strongest fitness advantage often dominate evolution. Thus, in the clinic, and in laboratory



**Figure 1**

**A multidrug treatment strategy that relies on all mutants having the same tradeoffs.**

(**A**) All of the mutants that resist Drug A do so via a similar mechanism such that all are sensitive to Drug B. (**B**) There are multiple different types of mutants that resist Drug A, not all of which are sensitive to Drug B.

experiments, the same drug-resistant mutations repeatedly emerge (Lupetti et al. 2002 [↗](#); Berkow and Lockhart 2017 [↗](#); Melnikov et al. 2020 [↗](#); Ksiezopolska et al. 2021 [↗](#)), potentially leading to the false impression that the mechanistic basis of resistance to a particular drug, and the associated tradeoffs, are less varied than may be true. This problem is amplified by the limitations of bulk DNA sequencing methods which often miss mutations that are present in less than 10% of a population's cells (Good et al. 2017 [↗](#)). A similar problem results from strategies to disentangle adaptive from passenger mutations that rely on observing the adaptive ones multiple times in multiple independent replicates (Martínez and Lang 2023 [↗](#)). In order to design better multidrug treatment strategies that thwart resistance, or to see if such strategies are feasible, we need methods to survey a more complete set of mutations and mechanisms that can contribute to resistance.

Fortunately, single-cell and single-lineage DNA sequencing technologies are allowing us to more deeply sample genetic diversity in evolving populations of microbes beyond the mutations that dominate evolution (Schmidt and Efferth 2016 [↗](#)). Here, we leverage a platform to perform massively-replicate evolution experiments in yeast. This platform has been shown to reveal a fuller spectrum of mutations underlying adaptation to a particular environment (Levy et al. 2015 [↗](#)). One key to its success is that it uses DNA barcodes to track all competing adaptive lineages, not just the ones that ultimately rise to appreciable frequency. Another key feature is that it captures adaptive lineages before they accumulate multiple mutations such that it is easy to pinpoint which mutation is adaptive. We apply this platform to investigate mechanisms underlying resistance to a specific antifungal drug: fluconazole (FLU) (Wang et al. 2022 [↗](#); Logan et al. 2022 [↗](#)). Although serious fungal infections are most common in immunocompromised individuals, their impact on global health is still striking, resulting in over 1.5 million deaths annually (Xie et al. 2014 [↗](#); Iyer et al. 2022 [↗](#)). By focusing on mechanisms of FLU resistance, we contribute to a growing literature about the tradeoffs that may be leveraged to design multidrug antifungal treatment strategies (Cowen and Lindquist 2005 [↗](#); Hill et al. 2015 [↗](#); Ksiezopolska et al. 2021 [↗](#); Iyer et al. 2022 [↗](#)). However, our primary goal is more generic: we seek to explore the utility of a high-throughput evolutionary approach to enumerate classes of drug resistant mutant and their associated tradeoffs.

To enhance the diversity of drug resistant mutants in our experiment, we performed multiple laboratory evolutions in a range of FLU concentrations and sometimes in combination with a second drug. We did so because previous work has shown that different drug concentrations and combinations select for different azole resistance mechanisms (Cowen and Lindquist 2005 [↗](#); Hill et al. 2015 [↗](#)). Ultimately, we obtained a large collection of 774 adaptive yeast strains. But how do we know whether we succeeded in isolating diverse types of FLU-resistant mutants? Typical phenotyping methods, *e.g.*, quantifying expression levels of drug export pumps (Miyazaki et al. 1998 [↗](#)) or of the drug targets themselves (Palmer and Kishony 2014 [↗](#)), are low throughput and require some a priori knowledge of the phenotypes that may be involved in drug resistance. Alternatively, many studies focus on identifying the genetic basis of adaptation in order to glean insights about the underlying mechanisms of resistance (Tenaillon et al. 2012 [↗](#); Cowen et al. 2014 [↗](#); Venkataram et al. 2016 [↗](#)). However, genotyping lineages from barcoded pools is technically challenging (Venkataram et al. 2016 [↗](#)), and further, genotype does not necessarily predict phenotype (Eguchi et al. 2019 [↗](#); Brettner et al. 2022b [↗](#)). For example, previous work using the same barcoded evolution platform used here discovered that the mutations that provide an advantage under glucose-limitation are in genes comprising a canonical glucose-sensing pathway (Venkataram et al. 2016 [↗](#)). Yet despite this similarity at the genetic level, follow-up work showed that these mutants did not experience the same tradeoffs when exposed to new environments (Li et al. 2018b [↗](#); Kinsler et al. 2020 [↗](#)).

Instead of trying to identify the phenotypic or even the genetic basis of adaptation, here we strive to enumerate different classes of FLU-resistant mutants. We sort evolved FLU-resistant yeast strains into classes based on whether they share similar tradeoffs across 12 different

environments. Previous work suggests that mutants with different fitness tradeoffs may affect fitness through different molecular mechanisms (Rodrigues et al. 2016 [↗](#); Li et al. 2019 [↗](#); Pinheiro et al. 2021 [↗](#)). Our work can thus be seen as part of a growing push to flip the problem of mechanism on its head (Li et al. 2019 [↗](#); Kinsler et al. 2020 [↗](#); Petti et al. 2023 [↗](#)). Instead of using a mechanistic understanding to predict a microbe's fitness, here we use how fitness varies across environments to enumerate classes of FLU-resistant mutant that likely affect fitness via different mechanisms. Across our collection of 774 adaptive yeast lineages, we discovered at least 6 distinct groups with characteristic tradeoffs. For example, we find some drug resistant mutants are generally advantageous, while others are advantageous only in specific environments. And we find some mutants that resist single drugs also resist combinations of those drugs, while others do not. By grouping mutants with similar tradeoffs, we reduce the number of unique drug-resistant mutants from more than can be easily phenotyped (774) to a manageable panel of 6 types for investigating the mechanistic basis of resistance.

With regard to multidrug regimens that exploit tradeoffs (**Figure 1** [↗](#)), our finding of multiple mutant classes with different tradeoffs suggests this may not be straightforward. The outlook is further complicated by our finding that some classes of FLU-resistant mutant primarily emerge from evolution experiments that did not contain FLU. This, as well as limits on our power to observe mutants with strong tradeoffs, suggest there may be additional types of FLU-resistant mutant beyond those we sampled. These observations suggest multidrug strategies that assume resistant mutants suffer consistent or common tradeoffs will often fail.

On the other hand, nuanced strategies to forestall resistance that allow for multiple mutant types are emerging (Maltas and Wood 2019 [↗](#); Gjini and Wood 2021 [↗](#); Wang et al. 2023 [↗](#)). For example, one idea is to apply a drug regimen that enriches for mutants that suffer a particular tradeoff before exploiting that tradeoff (Iram et al. 2020 [↗](#)). Another idea is to perform single-cell sequencing on infectious populations to discover which classes of mutants are present (Nagasawa et al. 2021 [↗](#); Forsyth et al. 2021 [↗](#)) and design treatments specific to those (Maltas and Wood 2019 [↗](#); Aissa et al. 2021 [↗](#); Hsieh et al. 2022 [↗](#)). Our findings support that such ideas may be feasible by demonstrating that there are not as many unique fitness tradeoffs as there are mutations.

More generally, our work – showing that 774 mutants fall into a much smaller number of groups – contributes to growing literature suggesting that the phenotypic basis of adaptation is not as diverse as the genetic basis (Kinsler et al. 2020 [↗](#); Iwasawa et al. 2022 [↗](#); Petti et al. 2023 [↗](#)). This winnowing of diversity is important: it may mean that evolutionary processes, for example, whether an infectious population will adapt to resist a drug, are sometimes predictable (Rodrigues et al. 2016 [↗](#); Lässig et al. 2017 [↗](#); Kinsler et al. 2020 [↗](#); Yoon et al. 2021 [↗](#); King et al. 2022 [↗](#); Wortel et al. 2023 [↗](#)).

## Results

### Barcoded evolution experiments uncover hundreds of yeast lineages with adaptive mutations

In order to create a sizable collection of drug-resistant mutants, we performed high-replicate evolution experiments utilizing barcoded yeast (Levy et al. 2015 [↗](#); Li et al. 2018b [↗](#); Boyer et al. 2021 [↗](#)). This barcoding system allows evolving hundreds of thousands of genetically identical yeast lineages together in a single flask. Each lineage is tagged with a unique DNA barcode, which is a 26 base pair sequence of DNA located within an artificial intron. Lineages with unique barcodes can be thought of as independent replicates of an evolution experiment. This high-replicate system has the potential to generate many different yeast lineages that differ by single adaptive mutations (Venkataram et al. 2016 [↗](#); Kinsler et al. 2020 [↗](#)).

We performed a total of 12 barcoded evolution experiments, each of which started from the same pool of approximately 300,000 barcoded yeast lineages (**Figure 2A** – **B**; **Figure S1**). These evolutions survey how yeast cells adapt to different concentrations and combinations of two drugs: fluconazole (FLU) and radicicol (RAD). FLU is a first line of defense against yeast infections, but over the past two decades diverse resistant mutations have been identified (Bongomin et al. 2017; Rybak et al. 2022; Osset-Trénor et al. 2023). Some earlier work suggested that FLU-resistant mutants are sensitive to the second drug we study, radicicol (RAD) (Cowen and Lindquist 2005; Cowen et al. 2009), and more generally that RAD can prevent the emergence of drug resistance in other systems (Whitesell et al. 2014). However, there are some mutants that are cross resistant to both FLU and RAD (Hill et al. 2015), and the prominent mechanism of resistance can differ with the intensity of selection and drug concentration (Cowen and Lindquist 2005; Yang et al. 2023). We thus chose to evolve yeast to resist different concentrations and combinations of FLU and RAD to generate a diverse pool of adaptive mutations comprising different mechanisms of drug resistance.

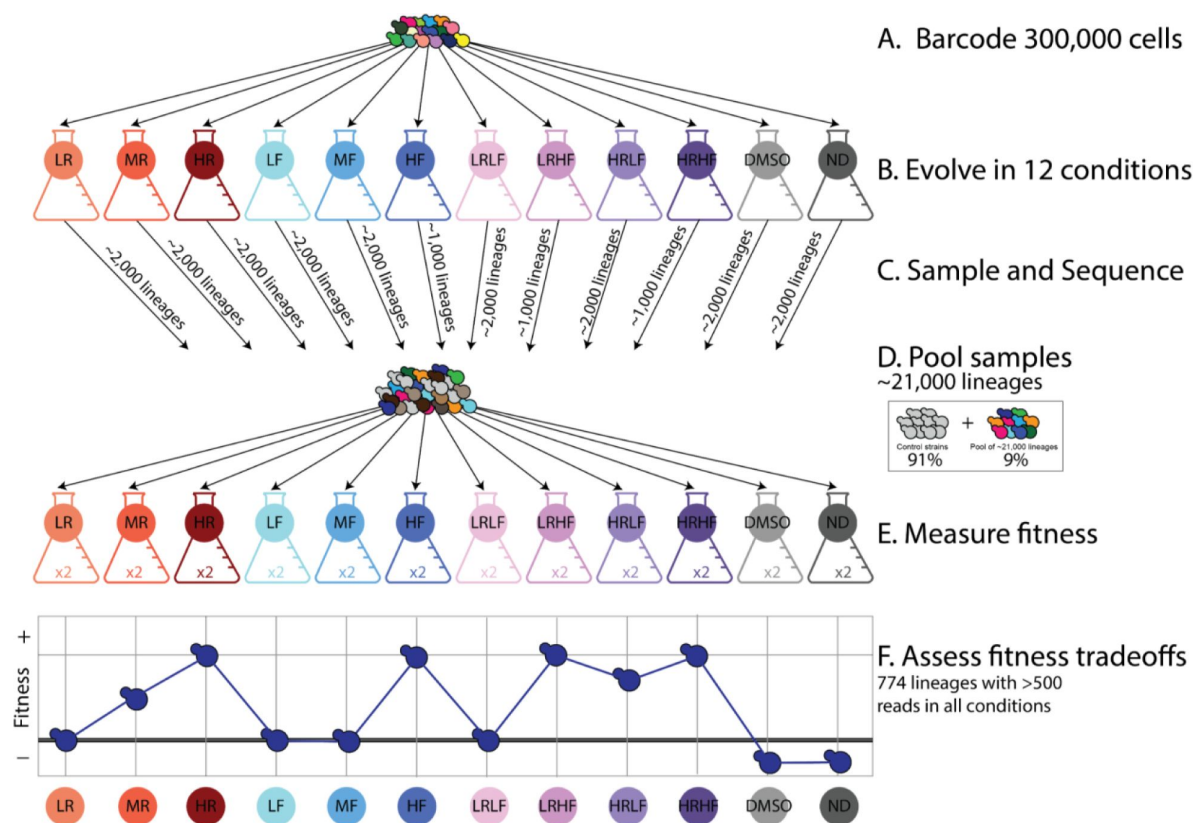
We evolved yeast to resist three different concentrations of either FLU and RAD for a total of six single-drug conditions (**Table 1**). We also studied four conditions containing combinations of both drugs, as well as two control conditions, for a total of 12 evolution experiments (**Table 1**). We chose to study subclinical drug concentrations with the hope that no drug treatment would be strong enough to reduce the population of yeast cells to only a handful of unique barcodes (**Figure S2**). We needed to maintain barcode diversity in order to evolve a large number of unique lineages that each accumulate different mutations.

With the goal of collecting adaptive lineages from each evolution experiment, we took samples from each one after 3 – 6 growth/transfer cycles (**Figure S1**). This represents roughly 24 to 48 generations of growth assuming 8 generations per growth/transfer cycle (Levy et al. 2015). We sampled early because previous work using this barcoded evolution system demonstrated that the diversity of adaptive lineages peaks after just a few dozen generations (Levy et al. 2015; Venkataram et al. 2016). We sampled about 2,000 cells from each evolution experiment except those 3 containing high concentrations of FLU from which we sampled only 1,000 cells for a total of ~21,000 isolates (2,000 cells x 9 conditions + 1000 cells x 3 conditions) (**Figure 2C**).

Next, we measured the fitness of each isolate in each of the 12 evolution environments to quantify fitness tradeoffs (e.g., whether mutants that perform well in one environment perform worse in another). This process also indirectly screens isolates for adaptive mutations by comparing the fitness of each evolved isolate to the ancestor of the evolution experiments (Venkataram et al. 2016). To do so, we pooled these 21,000 isolates and used this pool to initiate fitness competition experiments (**Figure 2D**). We competed the pool against control strains, i.e., strains of the ancestral genotype that do not possess adaptive mutations (Venkataram et al. 2016; Kinsler et al. 2020). We performed 24 such competitive fitness experiments, 2 per each of the original 12 evolution conditions (**Figure 2E**). In each experiment, we emulated the growth and transfer conditions of the original evolution experiments as precisely as possible, tracking how barcode frequencies changed over 5 growth/transfer cycles (~40 generations). We used the log-linear slope of this change, relative to the average slope for the control strains, to quantify relative fitness.

We found that many barcoded lineages have higher fitness than the control strains in some conditions, presumably because they possess adaptive mutations that improve their fitness in some conditions (**Figure S3**). In fact, some of these adaptive lineages outcompeted the other lineages so quickly that it posed a challenge. Barcodes pertaining to outcompeted lineages were often not present at high enough coverage to track their fitness. We applied a conservative filter, preserving only 774 lineages with barcodes that were observed >500 times in at least one replicate experiment per each of the 12 conditions. The reason we required fitness measurements in all 12





**Figure 2**

### An overview of the experimental design.

(A) Yeast cells were barcoded to create 300,000 lineages. (B) These lineages were evolved in 12 different conditions (Table 1). (C) A small sample of evolved lineages was taken from each evolution experiment and sequenced. (D) These samples of evolved lineages were all pooled together with control strains representing the ancestral genotype. (E) The fitness of each lineage in this pool of ~21,000 was measured in each of the 12 evolution conditions via pooled fitness competitions. Fitness was measured by tracking changes in barcode frequencies over time relative to control strains. Two replicates per condition were performed. (F) The overall goal is to investigate the fitness tradeoffs for each evolved lineage. For example, the lineage depicted in dark blue has high fitness in some environments (HR, HF) but lower fitness in others (DMSO, ND). We were able to investigate fitness tradeoffs for 774 of the 21,000 lineages. We excluded lineages when we did not observe their associated barcode at least 500 times in all 12 environments. In other words, we only included lineages for which we obtained high quality fitness estimates in all 12 environments.

| Evolution Condition                   | Abbreviation | SYMBOL                                                                                |
|---------------------------------------|--------------|---------------------------------------------------------------------------------------|
| 4 µg/ml Fluconazole                   | Low Flu      |    |
| 6 µg/ml Fluconazole                   | Med Flu      |    |
| 8 µg/ml Fluconazole                   | High Flu     |    |
| 5 µM Radicicol                        | Low Rad      |    |
| 15 µM Radicicol                       | Med Rad      |    |
| 20 µM Radicicol                       | High Rad     |    |
| 5 µM Radicicol + 4 µg/ml Fluconazole  | LRLF         |   |
| 5 µM Radicicol + 8 µg/ml Fluconazole  | LRHF         |  |
| 15 µM Radicicol + 4 µg/ml Fluconazole | HRLF         |  |
| 15 µM Radicicol + 8 µg/ml Fluconazole | HRHF         |  |
| 0.5% DMSO                             | DMSO         |  |
| No Drug                               | No Drug      |  |

**Table 1**

**A list of the environments included in this study and the symbol used to represent them in subsequent figures.**



conditions is that our goal is to examine each lineage's fitness tradeoffs (**Figure 2F**) to see if different lineages have different tradeoffs. In order to compare apples to apples, we need to measure each lineage's fitness in the same set of environments.

The 774 lineages we focus on are biased towards those that are reproducibly adaptive in multiple environments we study. This is because lineages that have low fitness in a particular environment are rarely observed >500 times in that environment (**Figure S4**). Despite this bias, we will go on to demonstrate that there are different types of mutants with characteristically different fitness tradeoffs present among these 774 lineages.

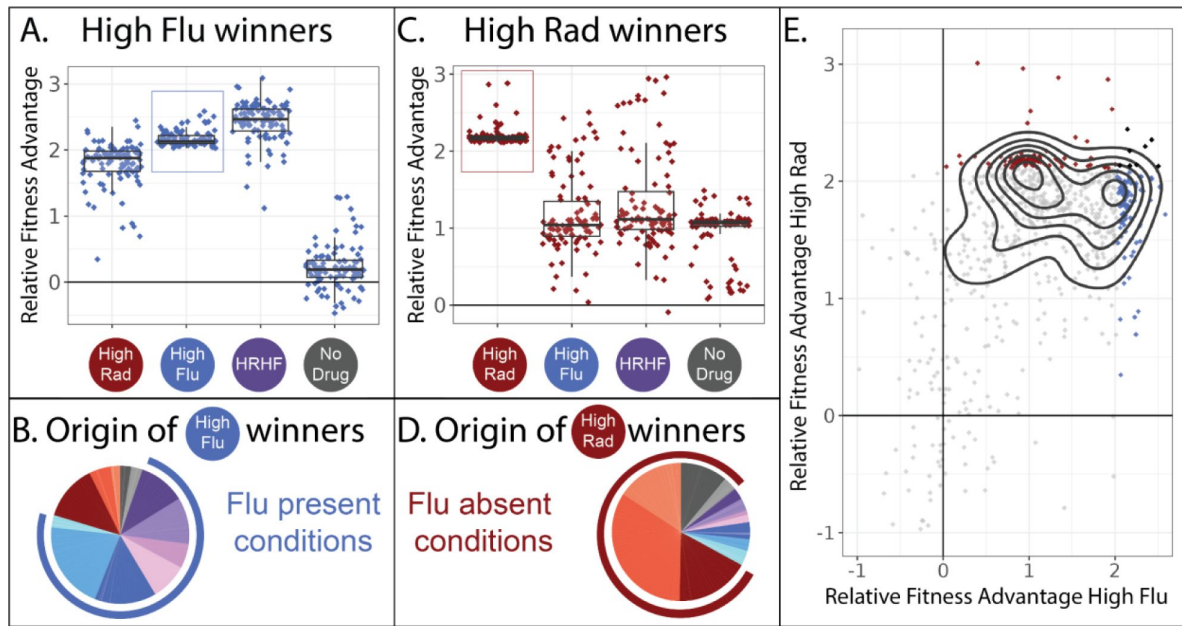
To provide additional evidence that these 774 barcoded yeast lineages indeed possess adaptive mutations, we performed whole genome sequencing on a subset of 62 strains. Because we sampled these lineages after only a few dozen generations of evolution, each lineage differs from the ancestor by just a few mutations, making it easy to pinpoint the genetic basis of adaptation. Doing so revealed mutations that have previously been shown to be adaptive in our evolution conditions (**Table S1**). For example, we sequenced many FLU-resistant yeast lineages finding 35 with unique single nucleotide mutations in either PDR1 or PDR3, and a few with mutations in SUR1 or UPC2, genes which have all been shown to contribute to FLU resistance in previous work (Flowers et al. 2012; Vasicek et al. 2014; Tanaka and Tani 2018; Uemura and Moriguchi 2022; Vu and Moye-Rowley 2022). Similarly, lineages that have very high fitness in RAD were found to possess single nucleotide mutations in genes associated with RAD resistance, such as HDA1 (Robbins et al. 2012) and HSC82, which is the target of RAD (Roe et al. 1999). We also observed several lineages with similar mutations to those observed in other studies using this barcoded evolution regime, including mutations to IRA1, IRA2 and GPB2 (Venkataram et al. 2016; Kinsler et al. 2020). Previous barcoded evolutions also observed that increases in ploidy were adaptive, with 43% to 60% of cells becoming diploid during the course of evolution (Venkataram et al. 2016). However, ploidy changes contributed less to adaptation in our experiment, with at most 9.4% of cells becoming diploid by the time point when we sampled, but often less than 2% (**Table S2**).

In sum, we have created a diverse pool of 774 barcoded yeast lineages, most of which have a fitness advantage in at least one of the conditions we study and are likely to possess a unique adaptive mutation. The question we address for the rest of this study is to what extent these hundreds of mutant lineages differ from one another in terms of their fitness tradeoffs and the mechanism/s underlying their fitness advantages.

## A unique mechanisms of FLU resistance emerges among mutants isolated in RAD evolutions

The majority of the 774 adaptive lineages that we study have higher fitness than the ancestral strains in not one, but often in several drug conditions. This suggests that pleiotropy, and in particular cross-resistance, is prevalent among the lineages we study. But not all lineages show the same patterns of cross resistance (**Figure 3**). For example, the 100 most fit lineages in our highest concentration of fluconazole are also beneficial in our highest concentration of radicicol (**Figure 3A**; leftmost two boxplots). As expected, these 100 lineages also have high fitness in conditions where high concentrations of FLU and RAD are combined (**Figure 3A**; third boxplot). And these 100 most-fit lineages in FLU lose their fitness advantage in conditions where no drug is present (**Figure 3A**; rightmost boxplot).

Given their high fitness in conditions containing FLU, it seems likely that these 100 mutants originated from evolution experiments containing FLU. We can trace every lineage back to the evolution experiment/s it originated from because we sequenced the lineages we sampled from each evolution experiment before pooling all 21,000 (**Figure 2C**). As we expected, these 100 best performing lineages in high FLU largely originate from evolution experiments containing FLU



**Figure 3**

### Two different classes of FLU-resistant mutants with unique tradeoffs.

(A) This panel describes the 100 mutant lineages with the highest fitness relative to the control strains in the high FLU environment (8 $\mu$ g/ml FLU). The vertical axis depicts the fitness advantages (log-linear slopes relative to control strains) for these 100 strains in four selected environments, including the high FLU environment (boxed). Boxplots summarize the distribution across all 100 lineages for each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times$  IQR (whiskers). (B) The 100 lineages with highest fitness in high FLU were most often sampled from evolution experiments in which FLU was present. In this pie-chart, colors correspond to the evolution conditions listed in [table 1](#) and the blue outer ring highlights evolution conditions that contain FLU. The size of each slice of pie represents the relative frequency with which these 100 lineages were found in each evolution experiment. (C) Similar to panel A, this panel describes the 100 mutant lineages with the highest fitness relative to the control strains in the high RAD environment (20 $\mu$ M Rad). (D) The 100 lineages with highest fitness in high RAD were most often sampled from evolution experiments that did not contain FLU. (E) A pairwise correlation plot showing that all 774 mutants, not just the two groups of 100 depicted in panels A and C, to some extent fall into two groups defined by their fitness in high FLU and high RAD. The contours (black curves) were generated using kernel density estimation with bins = 7. These contours describe the density of the underlying data, which is concentrated into two clusters defined by the two smallest black circles. The 100 mutants with highest fitness in high FLU are blue, highest fitness in high RAD are red, and the seven that overlap between the two aforementioned categories are black.

(**Figure 3B** [↗](#)). Given that these lineages have no fitness advantage in conditions containing no drug, it is also unsurprising that they are underrepresented in evolution experiments lacking RAD and FLU (**Figure 3B** [↗](#)).

It might be tempting to generalize that most mutations that provide drug resistance are not beneficial in environments without drugs. Afterall, we show this is true for 100 independent lineages (**Figure 3A** [↗](#)). Further, many previous studies find a similar pattern, whereby drug resistant mutants often do not have high fitness in the absence of drug ([Andersson and Hughes 2010](#) [↗](#); [Basra et al. 2018](#) [↗](#); [Allen et al. 2019](#) [↗](#); [Melnikov et al. 2020](#) [↗](#)), such that treatment strategies have emerged that cycle patients between drug and no drug states, albeit with mixed success ([Baker et al. 2018](#) [↗](#); [Raymond 2019](#) [↗](#); [Wang et al. 2019](#) [↗](#); [Algazi et al. 2020](#) [↗](#)). However, this type of generalization is not supported by our data. We find that drug resistance can sometimes come with an advantage, rather than a cost, in the absence of a drug (**Figure 3C** [↗](#)). The top 100 most fit mutants in our highest concentration of RAD provide a fitness advantage in high RAD, high FLU, as well as in environments with no drug (**Figure 3C** [↗](#)). These observations suggest that there are at least two different mechanisms by which to resist FLU that result in different tradeoffs in other environments (**Figure 3A** [↗](#) vs. **3C**).

Intriguingly, these FLU-resistant lineages that maintain their fitness advantage in the absence of drug (**Figure 3C** [↗](#)) mainly originate from evolution experiments performed in conditions lacking FLU (**Figure 3D** [↗](#)). This highlights how the potential mechanisms by which a microbe can resist a drug may be more varied than is often believed. Typically, one doesn't search for FLU-resistant mutants by evolving yeast to resist RAD. Thus, typical studies might miss this unique class of FLU-resistant mutants.

In sum, there appear to be at least two different types of mutants present among our collection of 774 adaptive yeast lineages. One group has almost equally high fitness in RAD and FLU but has no fitness advantage over the ancestral strain in conditions without either drug (**Figure 3A** [↗](#) & **3E**). Another group is defined by very high fitness in RAD, moderately high fitness in FLU and moderately high fitness in conditions without either drug (**Figure 3C** [↗](#) & **3E**). When comparing fitness in RAD vs. FLU across all 774 lineages, not only the top 100 best performing in each drug, we see some evidence that they largely fall into the two main categories highlighted in **figures 3A** [↗](#) and **3C** (**Figure 3E** [↗](#)). Thus, it might be tempting to conclude that there are two different types of FLU-resistant mutant in our dataset. However, sorting mutants into groups using a pairwise correlation plot (**Figure 3E** [↗](#)) excludes data from ten of our twelve environments.

## A strategy to differentiate classes of drug-resistant mutants with different tradeoffs

The observation of two distinct types of adaptive mutants (**Figure 3** [↗](#)) made us wonder whether there were additional unique types of FLU-resistant mutants with their own characteristic tradeoffs. This is difficult to tell by using pairwise correlation like that in **Figure 3E** [↗](#) because we are not studying pairs of conditions, as is somewhat common when looking for tradeoffs to leverage in multidrug therapies ([Scarborough et al. 2020](#) [↗](#); [Melnikov et al. 2020](#) [↗](#); [Ardell and Kryazhimskiy 2021](#) [↗](#); [Larkins-Ford et al. 2022](#) [↗](#)). Instead, we have collected fitness data from 12 conditions to yield a more comprehensive set of gene-by-environment interactions for each mutant. This type of data, describing how a particular genotype responds to environmental change, is sometimes called a 'reaction norm' and can inform quantitative genetic models of how selection operates in fluctuating environments ([Gomulkiewicz and Kirkpatrick 1992](#) [↗](#); [Ogbunugafor 2022](#) [↗](#)) and how much pleiotropy exists in nature ([Yadav et al. 2015](#) [↗](#)). More recent studies refer to the changing performance of a genotype across environments as a 'fitness profile' or in aggregate, a 'fitness seascape,' and suggest these type of dynamic measurements are the key to designing effective multi-drug treatments ([King et al. 2022](#) [↗](#)) and to predicting evolution ([Lässig et al. 2017](#) [↗](#); [Kinsler et al. 2020](#) [↗](#); [Cairns et al. 2022](#) [↗](#); [Iwasawa et al. 2022](#) [↗](#); [Chen et al.](#)

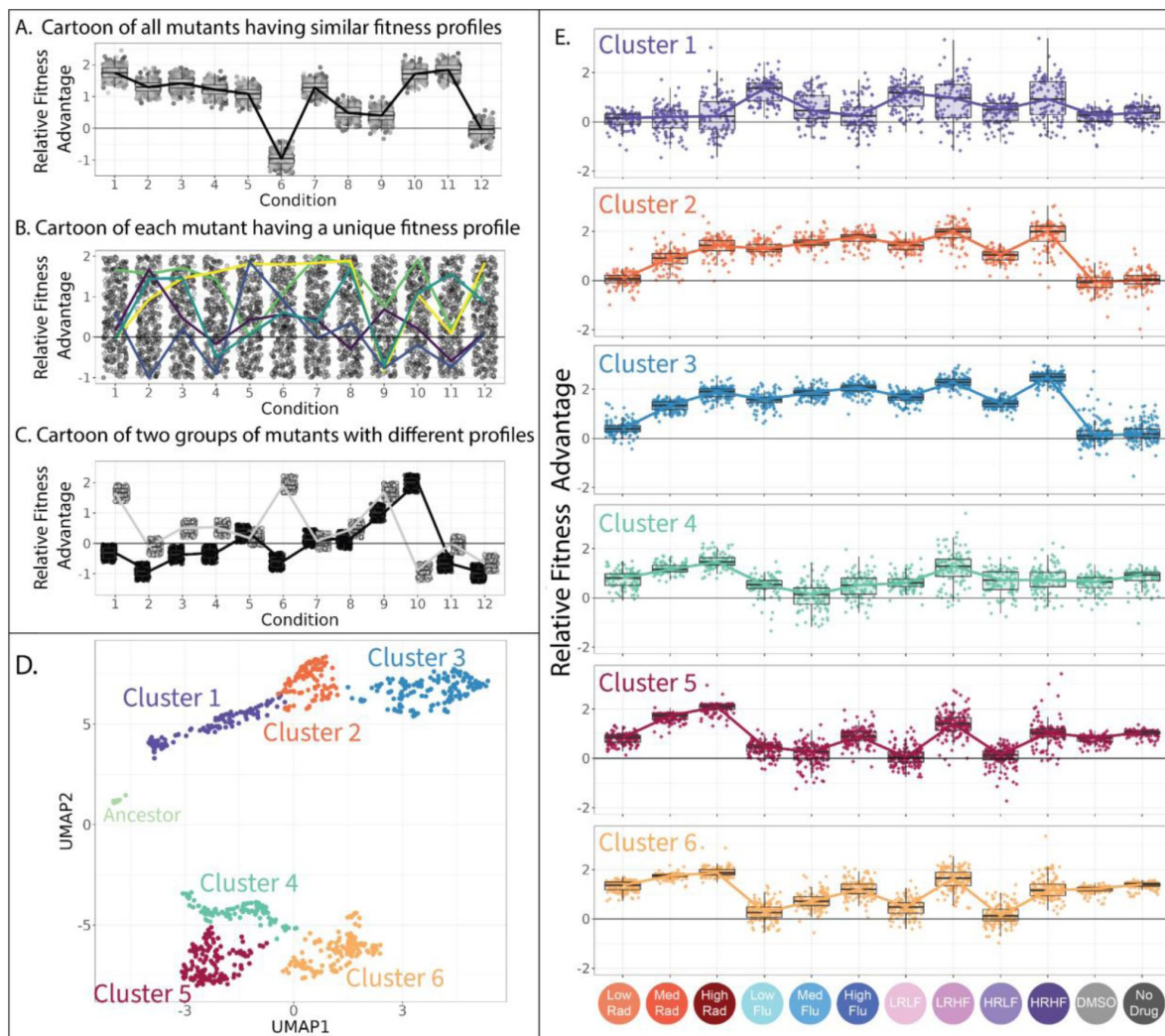
2023 [↗](#)). And when the environments studied represent different drugs, these types of data are often referred to as “collateral sensitivity profiles” a term chosen to convey how resistance to one drug can have “collateral” effects on performance in other drugs (Pál et al. 2015 [↗](#); Maltas and Wood 2019 [↗](#); Gjini and Wood 2021 [↗](#)). Despite the wide interest in this type of fitness data, it is technically challenging to generate, thus many previous studies of fitness profiles focus on a much smaller number of isolates (Imamovic et al. 2018 [↗](#); Nichol et al. 2019 [↗](#); Maltas and Wood 2019 [↗](#)), sometimes with variation restricted to a single gene (Mira et al. 2015 [↗](#); King et al. 2022 [↗](#)), or evolved in response to a single selection pressure (Li et al. 2018b [↗](#); Kinsler et al. 2020 [↗](#)). Here, we have generated fitness profiles for a large and diverse group of drug-resistant strains using the power of DNA barcodes. Now we seek to understand whether these mutants fall into distinct classes that each have characteristic fitness profiles (i.e., characteristic tradeoffs, characteristic reaction norms, or characteristic gene-by-environment interactions).

To address this question, we perform dimensional reduction, clustering mutants with fitness profiles that have a similar shape. It is in theory possible for all mutants to have similar profiles, perhaps implying they all affect fitness through similar underlying mechanisms (**Figure 4A** [↗](#)). However, the disparity reported in **Figure 3** [↗](#) suggests otherwise. It’s also possible that every mutant will have a different profile. This could happen if each mutant affects different molecular-level phenotypes that underlie its drug resistance (**Figure 4B** [↗](#)). But previous work suggests that the phenotypic basis of adaptation is less diverse than the genotypic basis. This is important because it means that evolutionary outcomes are more predictable at the level of phenotype (Kinsler et al. 2020 [↗](#); Brettner et al. 2022b [↗](#); Iwasawa et al. 2022 [↗](#)). Even so, every mutant could have a slightly different fitness profile if each affects the same handful of molecular-level phenotypes but to relatively different degrees. This would allow every mutant to have a unique response to environmental change, without requiring that there be as many unique molecular mechanisms underlying drug resistance as there are mutants. A final possibility is that there exist discrete classes of drug-resistant mutants with characteristic tradeoffs (**Figure 4C** [↗](#)). This might imply that each class of mutants provides drug resistance via a different molecular mechanism, or a different set of mechanisms. In sum, our endeavor to enumerate mutants with different fitness profiles speaks to general questions about the extent of pleiotropy in the genotype-phenotype-fitness map (Boyle et al. 2017 [↗](#); Geiler-Samerotte et al. 2020; Bakerlee et al. 2021 [↗](#); Chen et al. 2023 [↗](#)), the extent to which fitness tradeoffs are universal (Andersson and Hughes 2010 [↗](#); Li et al. 2019 [↗](#); Herren and Baym 2022 [↗](#)), and relatedly, the predictability of evolution and potential for controlling evolutionary processes (Lässig et al. 2017 [↗](#); Iram et al. 2020 [↗](#); Kinsler et al. 2020 [↗](#); King et al. 2022 [↗](#); Petti et al. 2023 [↗](#)).

To see whether there are distinct classes of adaptive mutants among our drug-resistant yeast lineages, we applied uniform manifold approximation and projection (UMAP) (McInnes et al. 2018a [↗](#)) to fitness measurements for 774 yeast strains across all 12 environments. This method places mutants with similar fitness profiles near each other in two-dimensional space. As might be expected, it largely places mutants in each of the two categories described in **Figure 3** [↗](#) far apart, with drug-resistant mutants that lose their benefit in the absence of drug in the top half of the graph, and those that maintain a benefit in the bottom half (**Figure 4D** [↗](#) and **Figure S5**).

Beyond the obvious divide between the top and bottom clusters of mutants on the UMAP, we used a gaussian mixture model (Fraley and Raftery 2003 [↗](#)) to identify clusters. A common problem in this type of analysis is the risk of dividing the data into clusters based on variation that represents measurement noise rather than reproducible differences between mutants (Zhao et al. 2008 [↗](#); Mirkin 2011 [↗](#)). One way we avoided this was by ceasing to split out additional clusters before doing so no longer improved model performance (**Figure S6**). The model we chose consists of seven clusters, including one pertaining to the control strains, and six others pertaining to different classes of adaptive mutant (**Figure 4D** [↗](#)). We investigated whether these clusters capture reproducible differences between mutants, rather than measurement noise, by reducing the amount of noise in our data and asking if the same clusters are still present. To do so, we





**Figure 4**

### Clustering evolved lineages with similar fitness profiles.

(A-C) Simulated data showing potential fitness profiles when (A) all mutants have similar responses to environmental change and thus a similar fitness profile, (B) every mutant has a different profile (five unique profiles are highlighted in color), or (C) every mutant has one of a small number of unique profiles (two unique profiles are depicted). (D) Every point in this plot represents one of the barcoded lineages colored by cluster; clusters were identified using a gaussian mixture model. The 774 adaptive lineages cluster into 6 groups based on variation in their fitness profiles; the control lineages cluster separately into the leftmost cluster in light green. (E) The fitness profiles of each cluster of adaptive lineages. Boxplots summarize the distribution across all lineages within each cluster in each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times \text{IQR}$  (whiskers).

reduced our collection of adaptive lineages from 774 to 617 by requiring 5,000 rather than 500 reads per lineage in order to infer fitness. This procedure reduced noise; the Pearson correlation across replicate experiments improved from 0.756% to 0.813%. Despite this reduction in variation, these 617 lineages cluster into the same six groups (plus a seventh pertaining to the control strains) as do the original 774 (**Figure S7**). The groupings are also preserved when we perform alternate methods for dimensionality reduction while adhering to a seven cluster model (**Figure S8**). Each of the six clusters of adaptive mutants that we identify has a characteristic fitness profile (**Figure 4E**). In any given environment, the fitness advantages of the mutants within each cluster are often very similar to one another and often significantly different from other clusters (**Figure 4E**). Our follow-up investigations provide additional evidence that these clusters of adaptive mutants each have characteristically different tradeoffs, suggesting they affect fitness via different molecular mechanisms.

## A group of mutants with distinct genotypes are primarily resistant to low concentrations of FLU

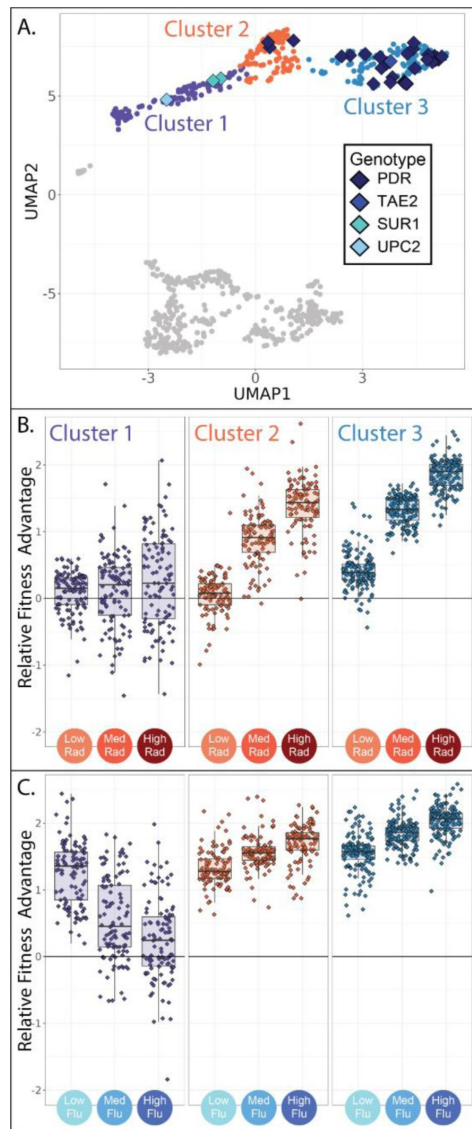
The upper three clusters of mutants on the UMAP (**Figure 4D**) are all similar in that they have elevated fitness in at least one FLU-containing environment but ancestor-like fitness in the absence of drug (**Figure 4E**; upper three profiles). Despite these similarities, there are major differences between these three groups of mutant lineages, both at the level of genotype and fitness profile (**Figure 5**). For example, in cluster 1 (depicted in purple in **Figures 4** and **5**), the 3 sequenced lineages have single nucleotide mutations to either SUR1 or UPC2 (**Figure 5A**). But in clusters 2 and 3 (depicted in blue and orange in **Figures 4** and **5**), 35/36 sequenced lineages have unique single nucleotide mutations to one of two genes associated with ‘Pleiotropic Drug Resistance’ (PDR1 or PDR3).

PDR1 and PDR3 are transcription factors that are well known to contribute to fluconazole resistance through increased transcription of a drug pump (PDR5) that removes FLU from cells (Fardeau et al. 2007; Osset-Trénor et al. 2023). However, SUR1 and UPC2 are less commonly mentioned in literature pertaining to FLU resistance, and have different functions within the cell as compared to PDR1 and PDR3 (Kapitzky et al. 2010; Hill et al. 2015). SUR1 converts inositol phosphorylceramide to mannosylinositol phosphorylceramide, which is a component of the plasma membrane (Uemura and Moriguchi 2022). Similarly, UPC2 is a transcription factor with a key role in activating the ergosterol biosynthesis genes, which contribute to membrane formation (Vik and Rine 2001; Tan et al. 2022). The presence of adaptive mutations in genes involved in membrane synthesis is consistent with fluconazole’s disruptive effect on membranes (Sorgo et al. 2011).

Interestingly, the lineages with mutations to UPC2 and SUR1, and the unsequenced lineages in the same cluster, do not consistently have cross resistance in RAD (**Figure 5B**; cluster 1). Oppositely, lineages with mutations to PDR1 or PDR3, and the unsequenced lineages in the same clusters, are uniformly cross resistant to RAD (**Figure 5B**; clusters 2 and 3). Perhaps this cross resistance is reflective of the fact that the drug efflux pump that PDR1/3 regulates (PDR5) can transport a wide range of drugs and molecules out of yeast cells (Kolaczowski et al. 1996; Harris et al. 2021). Overall, the targets of adaptation in cluster 1 have disparate functions within the cell as compared to the targets of adaptation in clusters 2 and 3. This may suggest that the mutants in cluster 1 confer FLU resistance via a different mechanism than clusters 2 and 3.

The lineages in cluster 1 have additional important differences from clusters 2 and 3. The lineages in cluster 1 perform best in the lowest concentration of FLU and have decreasing gains as the concentration of FLU rises (**Figure 5C**). In fact, about 15% of these mutant lineages perform worse than their ancestor in the highest concentration of FLU, suggesting the very mutations that provide resistance to low FLU are costly in higher concentrations of the same drug. The mutants in clusters 2 and 3 show the opposite trend from those in cluster 1: they perform best in the highest





**Figure 5**

**Evolved lineages comprising cluster 1 have different genotypes and phenotypes from neighboring clusters.**

(A) The three clusters on the top half of the UMAP differ in their targets of adaptation with cluster 1 being unique in that it does not contain mutations to PDR1 or PDR3. (B) Evolved lineages comprising cluster 1 do not have consistent fitness advantages in conditions containing RAD, while lineages comprising clusters 2 and 3 are uniformly adaptive in RAD. Boxplots summarize the distribution across all lineages within each cluster in each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times \text{IQR}$  (whiskers). (C) Lineages comprising cluster 1 are most fit in low concentrations of FLU, and this advantage dwindles as the FLU concentration increases. Lineages comprising clusters 2 and 3 show the opposite trend.

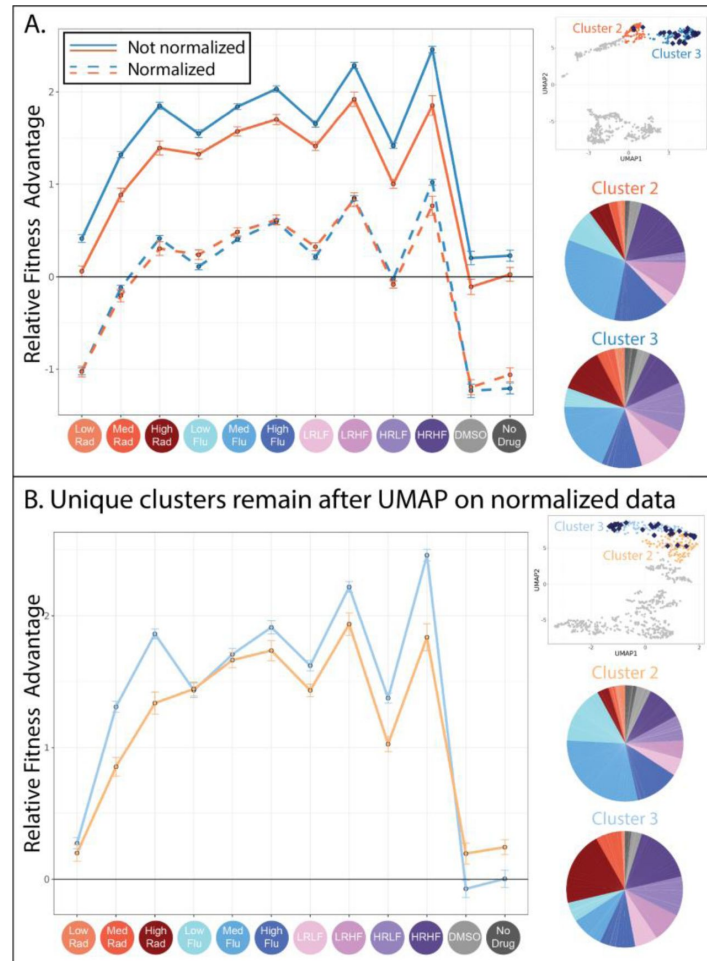
concentration of FLU and have reduced gains in lower concentrations (**Figure 5C**). These findings provide additional evidence that a distinct mechanism of FLU resistance distinguishes cluster 1 from clusters 2 and 3. The implication that different resistance mechanisms will dominate evolution in slightly different concentrations of the same drug highlights the complexity of adaptation and the need to more deeply understand the diversity of potential adaptive mechanisms before designing treatment strategies (Berman and Krysan 2020; Yang et al. 2023).

## Two groups of mutant lineages possessing similar adaptive mutations differ in sensitivity to RAD

While cluster 1 appears fairly different from its neighbors, it is not immediately obvious why the mutant lineages in clusters 2 and 3 are placed into separate groups. For one, the mutants in each cluster have fitness profiles with a very similar shape (**Figure 4E & 6A**). The sequenced lineages in each of these clusters also possess mutations to the same genes: PDR1 and PDR3 (**Figure 5A**). And finally, the lineages in each cluster originate from similar evolution experiments, largely those containing FLU (**Figure 6A**; pie charts). These observations made us wonder whether the difference between cluster 2 and 3 arose entirely because the mutants in cluster 3 have stronger effects than those in cluster 2 (**Figure 6A**; the solid blue line is above the solid orange line). In other words, we wondered whether the mutant lineages in clusters 2 and 3 affect fitness via the same mechanism, but to different degrees. To investigate this idea, we normalized all fitness profiles to have the same height on the vertical axis; this does not affect their shape (**Figure 6A**; dotted lines). Then we re-clustered and asked whether mutants pertaining to the original clusters 2 and 3 were now merged into a single cluster.

Normalizing in this way did not radically alter the UMAP, which still contains largely the same 6 clusters of mutants (**Figure S9**). Clusters 2 and 3, containing lineages with mutations to PDR1 or PDR 3, experienced the largest changes with 37% of mutants switching from one of these two groups to the other. The new clusters 2 and 3 now differ in the shape of their fitness profiles, whereby slight differences that existed between the original fitness profiles are exaggerated (**Figure 6B**). For example, mutants in cluster 3 perform better in high and medium concentrations of RAD (**Figure 6B**; line plot). This difference in fitness is reflected in the evolution experiments, with more mutant lineages in cluster 3 originating from the evolutions performed in RAD (**Figure 6B**; pie charts). Though cluster 3 mutants tend to have stronger RAD resistance, they tend to have reduced fitness in conditions containing neither FLU nor RAD as compared to cluster 2 lineages (**Figure 6B**; line plot). In sum, the differences between lineages in clusters 2 and 3 were not resolved upon normalizing fitness profiles to reduce magnitude differences, instead they were made more apparent (**Figure 6B**). These differences do not appear to be random because they persist across experiments. For example, cluster 3 mutants are more fit in both medium and high RAD environments (**Figure 6B**; line plot) and were more often isolated from evolutions containing RAD (**Figure 6B**; pie charts). These observations beg a question: how can different mutations to the same gene affect fitness via different molecular mechanisms?

Asking this question forces us to consider what we mean by “mechanism.” The mechanism by which mutations to PDR1 and PDR3 affect FLU resistance is well established: they increase transcription of an efflux pump that removes FLU from cells (Moye-Rowley 2019; Buechel and Pinkett 2020; Osset-Trénor et al. 2023). But if this is the only molecular-level effect of mutations to these genes, it is difficult to reconcile why PDR mutants fall into two distinct clusters with differently shaped fitness profiles. Others have also recently observed that mutants to PDR1 do not all behave the same way when exposed to novel drugs or changes in pH (Chen et al. 2023). This phenomenon is not reserved to PDR mutants, as adaptive missense mutations to another gene, IRA1, also do not share similarly shaped fitness profiles either (Kinsler et al. 2020). One explanation may be that, while all adaptive mutations within the same gene improve fitness



**Figure 6**

### Evolved lineages in clusters 2 and 3 have characteristic differences despite similarities at the genetic level.

(A) This panel shows the similarities between clusters 2 and 3. The upper right inset displays the same UMAP from **Figure 4D** with only clusters 2 and 3 highlighted and with lineages possessing mutations to the PDR genes depicted as blue diamonds. The line plot displays the same fitness profiles for clusters 2 and 3 as **Figure 4E**, plotting the average fitness for each cluster in each environment and a 95% confidence interval. Dotted lines represent the same data, normalized such that every lineage has an average fitness of 0 across all environments. These line plots show that the fitness profiles for clusters 2 and 3 have a very similar shape. Pie charts display the relative frequency with which lineages in clusters 2 and 3 were sampled from each of the 12 evolution conditions, colors match those in the horizontal axis of the line plot and **table 1**. (B) This panel shows the differences between the new clusters 2 and 3 created after all fitness profiles were normalized to eliminate magnitude differences. The upper right inset displays a new UMAP (also see **Figure S9**) that summarizes variation in fitness profiles after each profile was normalized by setting its average fitness across all environments to 0. The line plot displays the fitness profiles for the new clusters 2 and 3, which look different from those in panel A because 37% of mutants in the original clusters 2 and 3 switched identity from 2 to 3 or vice versa. The new clusters 2 and 3 are depicted in slightly different shades of blue and orange to reflect that these are not the same groupings as those depicted in **Figure 4**. Pie charts display the relative frequency with which lineages in new clusters 2 and 3 were sampled from each of the 12 evolution conditions, colors match those in the horizontal axis of the line plot and **table 1**.

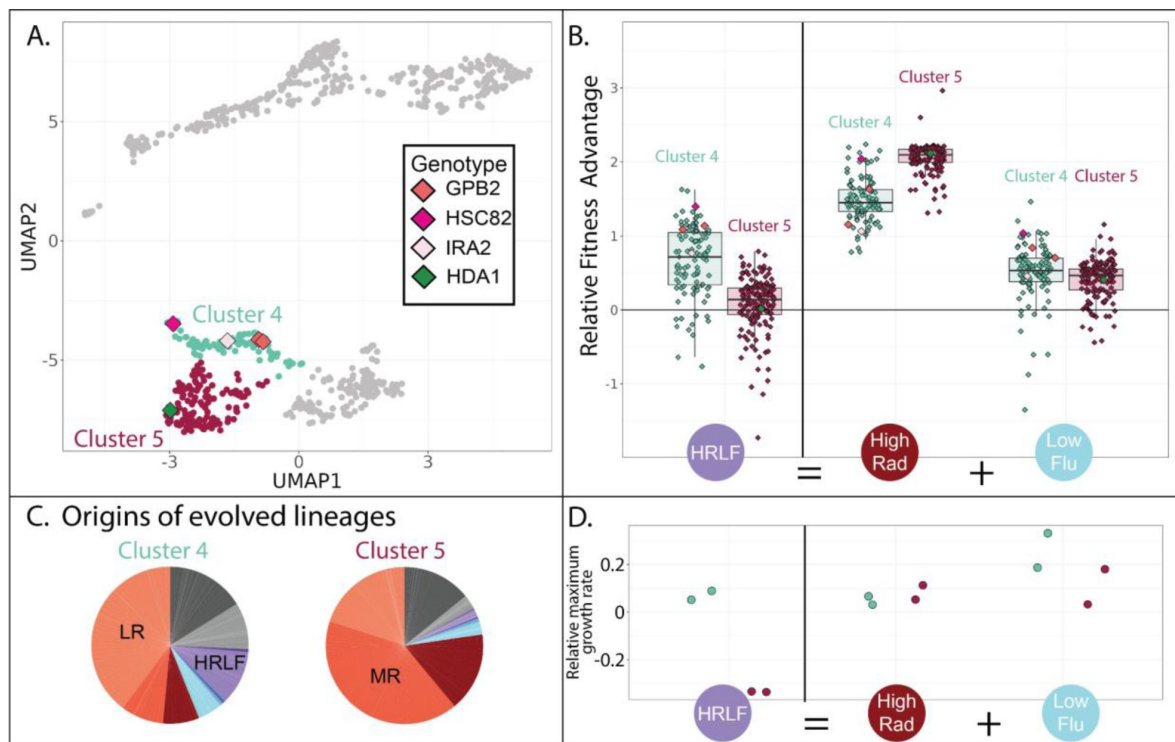
via the same mechanism, not all mutants suffer the same costs. For example, perhaps the adaptive PDR mutations in cluster 2 cause misfolding of the PDR protein, resulting in lower fitness in RAD because this drug inhibits a chaperone that helps proteins to fold. In this case, it might be more correct to say that each of our six clusters affects fitness through a different, but potentially overlapping, suite of mechanisms (Wang et al. 2023 [↗](#)). Previous work demonstrating that mutations commonly affect multiple traits supports this broader view of the mechanistic differences between clusters (Paaby and Rockman 2013 [↗](#); Boyle et al. 2017 [↗](#); Geiler-Samerotte et al. 2020 [↗](#); Kinsler et al. 2020 [↗](#)).

Alternatively, perhaps not all adaptive mutations to PDR improve fitness via the same mechanism. PDR1 and PDR3 regulate transcription of YOR1 and SNQ2 as well as PDR5, and maybe the different clusters we observe represent mutants that upregulate one of these downstream targets more than the other (Osset-Trénor et al. 2023). Or, the mutants in each cluster might harbor different aneuploidies or small, difficult to sequence chromosomal insertions or deletions that affect fitness. We leave identification of the precise mechanisms that differentiate these clusters for future work. Here, using the example of PDR mutants, we showcase how genotype may not predict fitness tradeoffs, suggesting there is more to learn about the mechanisms underlying FLU resistance.

## One group of RAD resistant mutants does not respond as expected to drug combinations

Though the three clusters of mutants on the bottom half of the UMAP are all advantageous in RAD and in conditions without any drug (Figure 4E [↗](#); lower three plots), they differ in their fitness in conditions containing FLU. For example, the cluster of yeast lineages highlighted in green (cluster 4 in Figures 4 [↗](#) and 7A [↗](#)) is unique in that it has a slight advantage in the HRLF environment (Figure 7B [↗](#)). We found it especially strange that the neighboring cluster 5 does not also have a fitness advantage in this condition. Mutants in cluster 5 have a slight advantage in the LF condition, and a big advantage in the HR condition, thus we expect them to have at least some fitness advantage in the condition where these two drugs are combined (HRLF), but they do not (Figure 7B [↗](#)). The same is true for the combination of LRLF: cluster 5 mutants have an advantage in both single drug conditions which is lost when the drugs are combined (Figure S10). However, the mutants in cluster 4 (green) exhibit no such sensitivity to combined treatment. They have a slight advantage in all of the aforementioned single drug conditions, which is preserved in the relevant multidrug conditions (Figure 7B [↗](#) & Figure S10). To obtain an independent measure of the fitness of cluster 4 vs. cluster 5 lineages in these multidrug conditions, we asked from where the lineages in each cluster originate. About 10% of cluster 4 lineages originated from the HRLF evolution, while almost none of the lineages in cluster 5 came from this experiment, confirming that cluster 5 lineages are uniquely sensitive to this multidrug environment (Figure 7C [↗](#)).

The different fitness profiles of mutants in cluster 4 versus 5 (Figure 7B [↗](#) & Figure S10) might imply that they have different growth phenotypes. We performed a follow-up experiment that supports this observation. We asked whether there are differences in the growth phenotypes of cluster 4 versus 5 mutants by measuring a growth curve for the lineage we were able to isolate and sequence from cluster 5, comparing it to a growth curve from a cluster 4 lineage (Figure 7D [↗](#)). Indeed, the selected mutants in cluster 4 and 5 appear to have markedly different growth curves in some conditions (Figure S10). The growth differences echo those we see in the fitness data. For example, the cluster 5 mutant has a lower maximum growth rate in the HRLF multidrug condition, corresponding with its lower fitness in this condition (Figure 7B [↗](#) & 7D). The cluster 5 mutant also reaches a lower maximum cell density in the LRLF multidrug condition and has lower fitness in this condition (Figure S10). However, the growth curves and fitness of cluster 4 and 5 mutants are more similar in LF, LR and HR single drug conditions. The observation of reproducible growth differences between a cluster 4 and cluster 5 mutant provides evidence that our clustering approach is effective at separating mutants with different properties.



**Figure 7**

### Evolved lineages in cluster 4 and 5 differ in response to combined drugs.

(A) Adjacent clusters 4 and 5 each contain a small number of sequenced isolates depicted as diamonds; diamond colors correspond to the genes containing adaptive mutations in each sequenced isolate. (B) Cluster 5 (red) has an unexpected fitness disadvantage in the HRLF multidrug environment relative to cluster 4 (green), given that cluster 5 lineages do not have a fitness disadvantage in the relevant single drug environments. Boxplots summarize the distribution across all lineages within each cluster in each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times \text{IQR}$  (whiskers). (C) Pie charts display the relative frequency with which lineages in each cluster were sampled from each of the 12 evolution conditions, colors match those in [table 1](#). (D) The maximum exponential growth rate for a single lineage isolated from each of clusters 4 (green) and 5 (red), relative to the ancestor. The growth rate of each lineage in each condition was measured twice by measuring changes in optical density over time. Tested lineage from cluster 4 (in green) has a mutation to GPB2 (S317T) while the lineage from cluster 5 (in red) has mutation to HDA1 (S600S).



## One group of RAD resistant mutants is exceptionally adaptive in conditions without drug

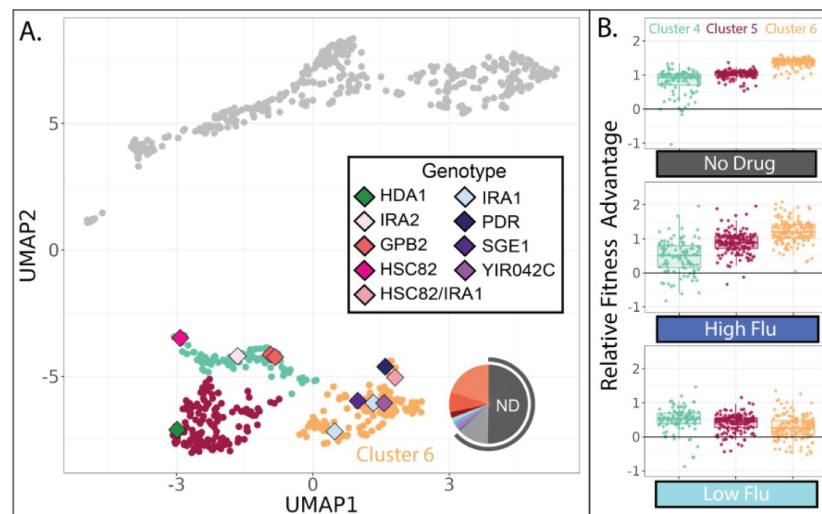
One group of mutants in the lower half of the UMAP (cluster 6 in [Figure 8A](#)) appears distinct from the other two in that it has the largest fitness advantage in conditions lacking any drug ([Figure 8B](#)). This might imply that cluster 6 lineages rose to high frequency during our evolution experiments in environments without either drug, specifically the “no drug” and “DMSO” control conditions. Indeed, this is what we observe: over 50% of the lineages in cluster 6 were sampled from one of these two evolution experiments ([Figure 8A](#)). On the contrary, the other clusters in the lower half of the UMAP consist mainly of lineages sampled from one of the RAD evolutions ([Figure 7C](#)). Since our fitness experiments were performed independently of the evolution experiments, this provides two independent pieces of evidence suggesting that lineages in cluster 6 are defined by their superior performance in conditions lacking any drug.

In line with the success of cluster 6 mutants in no drug conditions, the five sequenced mutants in this cluster include three that have mutations to IRA1, which was the most common target of adaptation in another evolution experiment in the condition we call “no drug” ([Figure 8A](#)) ([Venkataram et al. 2016](#)). In that experiment, and in our no drug experiment, mutations to IRA1 result in a greater fitness advantage than mutations to its paralog, IRA2, or mutations to other negative regulators of the RAS/PKA pathway such as GTP2 ([Figure 8](#)). Previous work showed that sometimes IRA1 mutants have very strong tradeoffs, for example, they become extremely maladaptive in environments containing salt or benomyl ([Kinsler et al. 2020](#)). We do not observe this to be the case for either FLU or RAD. In fact, we observe that cluster 6 mutants, including those in IRA1, maintain a fitness advantage in our highest concentration of both drugs ([Figure 4](#)), being more fit in high FLU than mutants in either of the other clusters in the lower half of the UMAP ([Figure 8B](#)). However, cluster 6 mutants are unique in that they lose their fitness advantage in the lowest concentration of FLU ([Figure 8B](#)). Being singularly sensitive to a low concentration of drug seems unusual, so much so that when this was observed previously for IRA1 mutants the authors added a note about the possibility of a technical error ([Kinsler et al. 2020](#)). Our results suggest that there is indeed something uniquely treacherous about the low fluconazole environment, at least for some genotypes.

## Discussion

Here, we present a barcoded collection of fluconazole (FLU) resistant yeast strains that is unique in its size, its diversity, and its tractability. One way we were able to isolate diverse types of FLU-resistance was by evolving yeast to resist diverse drug concentrations and combinations. But the more important tool used to increase both the number and type of mutants in our collection was DNA barcodes. These allowed us to sample beyond the drug resistant mutants that rise to appreciable frequency and to collect mutants that would eventually have been outcompeted by others. Our primary goal in collecting these mutants was to get a rough sense of how many different mechanisms of FLU resistance may exist. This question is relevant to evolutionary medicine (because more mechanisms of resistance make it harder to design strategies to avoid resistance), evolutionary theory (because more mechanisms of adaptation make it harder to predict how evolution will proceed), and genotype-phenotype mapping (because more mechanisms makes for a more complex map).

We distinguish mutants that likely act via different mechanisms by identifying those with different fitness tradeoffs across 12 environments, leveraging the mutants’ barcodes to track their relative fitness following previous work ([Kinsler et al. 2020](#)). The 774 FLU-resistant mutants studied here clustered into a handful of groups (6) with characteristic tradeoffs. Some groupings are unintuitive in that they segregate mutations within the same gene ([Figure 6](#)) or are distinguished by unexpectedly low fitness in multidrug conditions ([Figure 7](#)). These findings are important



**Figure 8**

**Evolved lineages in cluster 6 have higher fitness than other lineages in the absence of FLU and RAD.**

(A) Same UMAP as [Figure 4D](#) with clusters 4, 5, and 6 highlighted and sequenced isolates in these clusters represented as diamonds. Diamond colors correspond to the targets of adaptation in the sequenced isolates. Pie charts display the relative frequency with which lineages in cluster 6 were sampled from each of the 12 evolution conditions; colors match those in [table 1](#). (B) Of the three clusters on the bottom half of the UMAP, cluster 6 lineages perform best in conditions without any drug and with the highest concentration of FLU. Yet they perform worst in conditions with the lowest concentration of FLU. Boxplots summarize the distribution across all lineages within each cluster in each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times \text{IQR}$  (whiskers).

because they challenge strategies in evolutionary medicine that rely on consistent tradeoffs or intuitive trends when designing sequential drug treatments. On the other hand, the observation that some mutants have very similar tradeoffs such that they cluster together is promising in that it suggests predicting the impact of some mutations by understanding the impacts of others is somewhat feasible.

Problematically, the clusters we present are incomplete and bound to change as additional data presents itself. For one, we have shown that additional FLU-resistant mutants emerge from evolution experiments in conditions lacking FLU (**Figure 3C** & **3D**). This begs questions about what other FLU-resistant mutants might emerge in environments we have not studied here. Additionally, previous work has shown that some mutants that group together in our study (e.g., GPB2 and IRA2) have different fitness profiles in conditions that we did not include here (Kinsler et al. 2020). Also of note is that our evolution experiments were conducted for only a few generations and all started from the same genetic background. Additional types of FLU-resistant mutants with unique fitness profiles may emerge from other genetic backgrounds or arise after more mutations are allowed to accumulate (Brandis et al. 2012; Bosch et al. 2021; Allen et al. 2021). Finally, by requiring that all included mutants have sufficient sequencing coverage in all 12 environments, our study is underpowered to detect adaptive lineages that have low fitness in any of the 12 environments. This is bound to exclude large numbers of adaptive mutants. For example, previous work has shown some FLU resistant mutants have strong tradeoffs in RAD (Cowen and Lindquist 2005). Perhaps we are unable to detect these mutants because their barcodes are at too low a frequency in RAD environments, thus they are excluded from our collection of 774. All of the aforementioned observations combined suggest that there are more unique types of FLU-resistant mutations than those represented by these 6 clusters, and that the molecular mechanisms that can contribute to fitness in FLU are more diverse than we know.

On the up side for evolutionary medicine, not every infection harbors all possible types of mutants. This might explain why strategies that exploit one or two common tradeoffs have some, albeit mixed, success in delaying or preventing the emergence of resistance (Amin et al. 2015; Kaiser 2017; Imamovic et al. 2018; Wang et al. 2019; Krishna et al. 2022; Nyhoejen and Uecker 2023; Waller et al. 2023). Our results encourage more complex strategies to thwart drug resistance (Iram et al. 2020), such as those that focus on advance screening to determine the resistance mechanisms that are present (Andersson et al. 2019a), or on cycling a larger number of drugs to exploit a larger number of tradeoffs (Yoshida et al. 2017; Thomas et al. 2022). Problematically, these strategies often rely on knowledge about the diversity of mutants and tradeoffs that exist (or that can emerge) within an infectious population. This type of information about population heterogeneity, heteroresistance, and substructure is expensive and arduous to obtain (Andersson et al. 2019b; Bottery et al. 2021). Fortunately, new methods, in addition to the one presented in this study, are emerging (Kuchina et al. 2021; Aissa et al. 2021; Nagasawa et al. 2021; Forsyth et al. 2021; Hsieh et al. 2022; Brettner et al. 2022a). The richer data provided by these methods dovetails with emerging population genetic models that predict the likelihood of resistance to a given drug regimen (Read and Huijben 2009; Day et al. 2015; Wilson et al. 2016; Cannataro et al. 2018; Somarelli et al. 2020; Feder et al. 2021; King et al. 2022). In sum, our observation of numerous different types of drug-resistant mutations suggests that designing resistance-detering therapies is challenging, but perhaps not impossible.

Outside of predicting the evolution of resistance, our findings provide a tool to investigate the phenotypic impacts of mutation. This task has proven daunting in light of work demonstrating that mutations often have many phenotypic impacts (Paaby and Rockman 2013; Boyle et al. 2017) and that these impacts change with contexts including the environment (Paaby et al. 2015; Geiler-Samerotte et al. 2016, 2020; Lee et al. 2019; Eguchi et al. 2019). The approach presented in this study provides a way forward, not only by identifying mutations that cluster together such that the effects of some can be predicted from others, but also by identifying

environments that differentiate one cluster of mutants from another. This suggests where to look to understand the phenotypes that differentiate each cluster of mutants. For example, we were able to show that the growth phenotypes of mutants from clusters 4 and 5 are different because we knew to look for these differences in multidrug environments (**Figure 7**). And our results suggest radicicol environments may be most helpful in teasing out any phenotypic differences that set apart some PDR mutations from others (**Figure 6**). Thus, our approach guides efforts to understand the phenotypic effects of mutation, while also guiding efforts to predict the effects of some mutations from others as well as efforts to predict the outcomes of adaptive evolution.

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## Methods

### Base yeast strains

All of the yeast lineages studied here originated from the same starting strain referred to as the “landing pad strain” (SHA185) in previous work (Levy et al. 2015). We transformed a barcode library (generously provided by Sasha Levy) into this strain as described below, creating a strain with the following genetic background: MATa, ura3Δ0, ybr209w::Gal-Cre-KanMX-1/2URA3-loxP-Barcode-1/2URA3-HygMX-lox66/71.

### Base media

All experiments were conducted in “M3” media defined in the same study as the landing pad strain (Levy et al. 2015), which is a glucose-limited media lacking uracil. In our study, we supplemented this media with fluconazole, radicicol, or DMSO when appropriate.

### Selecting drug concentrations

Our goal was to choose concentrations of each drug that would not kill so many yeast cells as to dramatically decrease barcode diversity. We wanted to maintain a high number of unique barcodes so we could track a high number of yeast lineages as they independently evolved drug resistance. We measured the effect of each drug and drug combination on the growth rate of a single barcoded yeast strain using a plate reader to track changes in optical density (OD) over time. Ultimately we chose a “low” concentration of each drug that appeared to have no effect on growth rate, and a “high” concentration that appeared to reduce growth rate by about 15% (**Figure S2**). Though the lowest concentration of radicicol that we tested on a plate reader was 10 μM, we chose 5 μM as our low RAD concentration because previous work suggested this concentration had widespread effects on yeast physiology without affecting growth (Jarosz and Lindquist 2010; Geiler-Samerotte et al. 2016). To perform our plate reader experiment, a single colony was

grown to saturation. From this culture, 5  $\mu$ l was added to every well of a 96-well plate, where every well contained 195  $\mu$ l of M3 media. Some wells also contained either fluconazole, radicicol, DMSO, or combinations of these drugs. The concentrations that were tested are listed on the horizontal axis of **figure S2**; each drug condition was replicated six times. The 96-well plate was incubated at 30°C for 48 hours on a plate reader and OD measurements were taken every 30 minutes. Raw OD values were exported and maximum exponential growth rates for all tested conditions were calculated from the log-linear changes in OD over time.

## Inserting 300,000 unique DNA barcodes into otherwise genetically identical yeast cells

In order to track many yeast lineages as they independently develop drug resistance, we needed to insert unique DNA barcodes into many yeast cells. Plasmids harboring barcodes (pBar3) were the same as those used in a previous barcoded evolution experiment (Levy et al. 2015 [↗](#)) and were generously provided to us by Sasha Levy. These barcodes are 25 base pairs in length. They are targeted to an artificial intron within the Ura3 gene, such that they must be retained in media lacking uracil but are not expressed and thus do not themselves affect fitness (Levy et al. 2015 [↗](#)). We transformed this barcode library (pBar3) into the landing pad strain (SHA185) as was done previously, activating a Cre-lox recombination system by growing the cells in YP-galactose, which resulted in genomic integration of the barcode. However, our efforts to perform extremely high efficiency transformations from which we could isolate hundreds of thousands of uniquely barcoded yeast were unsuccessful, despite manipulating the levels and timing of the inducer (galactose). Ultimately we performed 24 separate transformations and pooled many of these to obtain a large pool of barcoded yeast where every yeast cell was genetically identical except for its DNA barcode.

## Examining the frequency of each barcode in the starting pool of cells

We sequenced the barcode region of these 24 transformed yeast populations on the Hiseq X platform using a dual index system (Kinsler et al. 2023 [↗](#)) to discern barcode coverage, i.e., how many total unique barcodes were successfully inserted into yeast cells and how evenly these barcodes were sampled. We needed many uniquely barcoded yeast in order to observe many different adaptive lineages within each evolution experiment. But barcodes with very high frequencies, referred to herein as monster lineages, were present in 10 of the 24 transformations and present a problem. Monster lineages allow too many cells to carry the same barcode, giving that barcode more chances to develop an adaptive mutation. This could allow different cells harboring that same barcode to pick up different adaptive mutations, destroying our ability to draw conclusions about adaptive mutations by using barcodes. Therefore, our final library of barcoded lineages was created by pooling 14 individual transformations together, choosing those 14 that lacked monster lineages, which we defined as lineages representing greater than 1% of all transformants. Our sequencing results suggest that this library contains about 300,000 unique barcodes.

## Initiating 12 barcoded evolution experiments

All evolution experiments started from the same pool of roughly 300,000 uniquely barcoded yeast lineages. To start the evolution experiments, a pea sized amount of the frozen yeast barcode library was grown up in 4 ml YPD for 4 hours at 30 °C in a shaking incubator at 220 rpm. Then, 300  $\mu$ l of the grown barcode library was added to each of 12 pre-prepared 500mL flasks representing the 12 evolution experiments listed in **Table 1** [↗](#). To prepare these flasks, first, 1.2 L of M3 media was warmed at 30°C. Then, 100 ml was added to each of 12 flat bottom flasks. Next, 500  $\mu$ l of the appropriate drug or drug combination was added to each flask. Drugs were pre-diluted, aliquoted



and frozen such that 500  $\mu$ l of the appropriate tube could be added to each flask to achieve the desired concentration as listed in **Table 1**. All drugs were resuspended in DMSO such that the final concentration of DMSO in all experiments (except the “no drug” control) was 0.5%.

## Performing barcoded evolution experiments

Evolution experiments were performed following previous work (Levy et al. 2015). After initiation (see above), the yeast in every flask were allowed to grow at 30°C with shaking at 200 RPM for 48 hours. Then, the flasks were removed from the incubator and 400 – 1000  $\mu$ l of each culture was transferred to a new pre-prepared flask with identical conditions to the first. The reason we added more volume (1000  $\mu$ l) to some flasks than previous work was that the cell counts at the end of the 48 hours were lower for some of our higher drug conditions. We adjusted the transfer volume to maintain a transfer population of  $4 \times 10^7$  cells, which was the same as in previous work (Levy et al. 2015). We completed a total of 24 growth/transfer cycles, corresponding to 192 generations of growth assuming 8 generations per 48-hour cycle (Levy et al. 2015). Following each transfer, the remaining culture from each flask were split into two 50 ml conical vials, centrifuged for 3 minutes at 4000 rpm, and the supernatant was discarded. The final pellet was resuspended in 30% glycerol up to a total volume of 6 ml before being split into three 2 ml cryovials and stored at –80°C. These frozen samples were later utilized for barcode sequencing and isolating adaptive mutants.

## Isolating a large pool of adaptive mutants

To generate a large pool of diverse adaptive mutants, our goal was to collect a sample from each evolution experiment at a time point when there were many different adaptive lineages competing. If we sampled too late, the adaptive lineage with the greatest fitness advantage would have already risen to high frequency, thus reducing diversity. But if we sampled too early, adaptive lineages would not yet have risen in frequency above other lineages. Therefore, we chose to sample cells from a time in each evolution experiment when many barcoded lineages appeared to be rising in frequency (**Figure S1**). We sampled either 1 or 2 thousand cells per each evolution experiment by spreading frozen stock from the chosen time point onto agarose plates, scraping 1 or 2 thousand colonies into a 15mL conical tube containing a final concentration of 30% glycerol, and freezing the pool pertaining to each of the 12 evolutions. We sampled 2,000 cells from most evolution experiments, but sampled only 1,000 from those containing a high concentration of FLU as those evolutions appeared to have reduced barcode diversity (**Figure S1**), presumably because high FLU represents a strong selective pressure. We sequenced the barcodes from each of these 12 pools so that we could track which adaptive mutants originated from which evolution experiment (see methods section below entitled, “**Inferring where adaptive lineages originally evolved**”).

## Initiating barcoded fitness competition experiments

To assess the fitnesses of the 1 or 2 thousand barcoded lineages that we sampled from each evolution experiment, we pooled all sampled lineages together into a larger pool of roughly 21,000 barcoded lineages. We used this larger pool to initiate 24 fitness competition experiments, 2 replicates for each of the 12 conditions listed in **Table 1**. In this type of competition, we measure fitness by tracking changes in each barcode’s frequency over time. Barcodes that rise in frequency represent strains that have higher fitness than others.

Our goal was to calculate the fitness effect of adaptive mutations. Therefore, we needed to calculate the fitness of every evolved lineage relative to the unmutated ancestor of the evolution experiments. To do so, we followed previous work by spiking in a large quantity of this unmutated ancestor strain into each fitness competition, with this ancestor making up at least 90% of the final culture (Venkataram et al. 2016; Kinsler et al. 2020). In environments containing a high concentration of FLU which resulted in the ancestral strain having a more severe growth defect, we spiked in the ancestor such that it represented 95% of the final pool.

To avoid wasting 90% or more of our sequencing reads on the ancestor strain's barcode, we created a barcodeless ancestor strain. This strain was created by transforming SHA185 with a linear piece of DNA such that the genetic background was identical to the strains of the barcoded library, but the homology to the primers used to amplify the barcode was missing. Thus the DNA from these cells does not get amplified or sequenced during subsequent steps.

In addition to this barcodeless ancestor, we also spiked in some barcoded ancestral strains at lower frequency (1%) to use as “reference” or “control” strains, following previous work (Kinsler et al. 2020 [↗](#), 2023 [↗](#)). These strains have been previously shown to possess no fitness differences from the ancestor. We used these strains as a baseline when calculating relative fitness by setting the fitness of these strains to zero during our fitness inference procedure (see methods section below entitled, “**Inferring fitness**”).

All 24 fitness competitions were performed simultaneously in one big batch (Kinsler et al. 2023 [↗](#)) and initiated from the same pool of roughly 21,000 barcoded evolved yeast lineages, barcodeless ancestor, and control strains. To initiate the competitions,  $7 \times 10^7$  cells from this pool were added to 24 pre-prepared 500mL flasks corresponding to the conditions listed in **Table 1** [↗](#). These flasks were prepared exactly the same way as was done for the evolution experiments (see above in “**Performing barcoded evolution experiments**”). Each flask was allowed to grow for 48 hours at 30°C with shaking at 200 RPM.

## Performing barcoded fitness competition experiments

Fitness competitions were performed following previous work (Kinsler et al. 2020 [↗](#)). After the initial flasks were allowed to grow for 48 hours, they were removed from the incubator and 400  $\mu$ l from each culture representing  $4 \times 10^7$  cells were transferred to a new flask with identical media. For each of 24 competitions, we completed a total of 4 growth/transfer cycles, corresponding to 40 generations of growth assuming 8 generations per 48-hour cycle (Levy et al. 2015 [↗](#)). Following each transfer, the remaining culture from each flask was split into two 50 ml conical vials, centrifuged for 3 minutes at 4000 rpm, and the supernatant was discarded. The final pellet was resuspended in 30% glycerol up to a total volume of 6 ml before being split into three 2 ml cryovials and stored at –80°C. These frozen samples were later utilized for DNA extraction and subsequent barcode sequencing.

## Extracting genomic DNA

DNA was extracted from 500  $\mu$ l of concentrated frozen stocks pertaining to the evolution experiments and fitness competitions. Frozen cells were thawed and pelleted. Cells were treated with 250  $\mu$ l of 0.1 M Na2EDTA, 1M sorbitol and 5U/ $\mu$ l zymolyase for a minimum of 15 minutes at 37 °C to remove the cell wall. Lysis was completed by adding 250  $\mu$ l of 1% SDS, 0.2N NaOH and inverting to mix. Proteins and cell debris were removed with 5M KOAc by spinning for 5 min at 15,000 rpm. Supernatant was moved to a new tube and DNA was precipitated with 600  $\mu$ l isopropanol by spinning for 5 min at 15,000 rpm. The resulting pellet was washed 1 ml of 70% ethanol before being resuspended in 50  $\mu$ l water plus 10 $\mu$ g/ml RNase. Extracted DNA was quantified using the NanoDrop spectrophotometer and all samples were diluted to a concentration of 50 ng/ $\mu$ L for barcode amplification and sequencing library preparation.

## Preparing barcodes for high-throughput multiplexed sequencing using PCR

Extracted DNA was prepared for sequencing using a two-step PCR that preserves information about the relative frequency of each barcode in each sample (Venkataram et al. 2016 [↗](#); Kinsler et al. 2020 [↗](#), 2023 [↗](#)). Briefly, in the first step PCR, the barcode region is amplified from the genomic DNA, labeled with a sample-specific combination of primers, and tagged with a UMI. This step utilizes a short 3 cycle PCR with New England Biolabs OneTaq polymerase. Purification of the first

step product to remove excess reagents was performed using Thermo Scientific GeneJET PCR Purification Kit. The second step PCR attached Illumina indices that were used to distinguish samples from different experiments and timepoints. We utilized a dual indexing scheme to prevent index misassignment that is common when sequencing amplicon libraries using patterned flow cell technology (Kinsler et al. 2023 [↗](#)). Amplification of this second step of PCR was done with a longer 23 cycle PCR using Q5 polymerase. Final libraries were bead purified using 0.8X Quantabio sparQ Pure Mag beads. Quantification of the final PCR products was done using the Invitrogen Qubit Fluorometer before all samples were pooled at equimolar ratios for sequencing.

## Sequencing and clustering barcodes

Next Generation Sequencing was performed at either Psomagen (Rockville, MD) or at the Translation Genomics Research Institute (Phoenix, AZ) on patterned flow cells (either an Illumina HiSeqX or NovaSeq) using 2 x 150 base pair paired end reads. Samples were dual indexed to allow multiplexing while minimizing contamination from index misassignments (Kinsler et al. 2023 [↗](#)). The 20 base pairs of variable sequence referred to as a DNA barcode were identified and clustered to determine the number of unique barcodes and the frequency of each barcode in each sample. For the evolution experiments, this was done following our previous work (Venkataram et al. 2016 [↗](#); Kinsler et al. 2020 [↗](#)). For the fitness competition experiments, this was done using updated software (Zhao et al. 2018 [↗](#)) with the following command: `bartender_extractor_com -f ${SAMPLE}_R1_001.fastq -o ${SAMPLE}_extracted -q 0 -p GTACC[5]AA[5]AA[5]TT[5]ATAAC -m 2 -d f -u 0,8`

## Inferring fitness

In fitness competition experiments, fitness is often inferred from the log-linear change in a strain's frequency over time (Geiler-Samerotte et al. 2011 [↗](#); Bakerlee et al. 2021 [↗](#); Kinsler et al. 2023 [↗](#)). Recently, more advanced methods to infer fitness have emerged that take into account nonlinearities in frequency changes over time, for example, nonlinearities that reflect changes in the mean fitness of the population (Venkataram et al. 2016 [↗](#); Li et al. 2018a [↗](#), 2023 [↗](#); Kinsler et al. 2020 [↗](#)). We had trouble implementing these newer methods on our fitness data, perhaps because many of our evolved lineages, and our control strains, have low fitness in some drugs. This caused their barcodes to rapidly decline in frequency such that they received low counts only at later time points. Their counts could become so low that these lineages would seemingly disappear due to sampling error, and then reappear at a subsequent time point. This dramatic (but false) late increase in frequency was sometimes interpreted as evidence of very high fitness, especially when we inferred fitness using approaches that account for nonlinearities.

To contend with this issue, we applied strict coverage thresholds to every fitness measurement: we required at least 500 counts across all timepoints in order to infer fitness for a given lineage in a given environment. This is stricter than previous work that does not require a minimum number of reads per each lineage and instead requires a minimum number of reads per time point (Kinsler et al. 2020 [↗](#)). We found that 774 lineages passed our threshold in at least one replicate experiment per all 12 environments. Of these, 729 passed for both replicates and the final fitness value we report represents the average of both replicates.

Even with our strict coverage threshold, fitness inference methods that account for nonlinearities still interpreted minor stochastic fluctuations in fitness at later time points as evidence of a fitness advantage, even if fitness dramatically declined in earlier time points. Therefore, we calculated fitness via the traditional method, as the slope of the log-linear change in barcode frequency relative to the average slope of the control strains. We found that this method is less sensitive to that type of error. Using this method, we found that our fitness inferences were reproducible between replicates (**Figure S4A**), and between experiments performed in similar conditions (e.g., medium vs. high concentrations of the same drug) (**Figure S4B**). When we increased our coverage

threshold to require an order of magnitude more reads per lineage per measurement (from 500 to 5000), we lost 157 lineages (from 774 to 617), saw reproducibility increase across replicates (from an average Pearson correlation of 0.756 to 0.813) and the main conclusions of our study were unchanged in that the same 6 clusters were present on a UMAP (**Figure S7**).

## Identifying adaptive mutations using whole-genome sequencing

One downside of barcoded evolution experiments is that all lineages exist together in a pooled culture. Fishing out adaptive lineages in order to perform whole genome sequencing is a major challenge (Venkataram et al. 2016 [↗](#)). Here, we randomly selected cells from these mixed pools for whole genome sequencing, sometimes selecting from later time points in the evolution experiments and sometimes selecting from the samples of 1 or 2 thousand cells that were isolated to initiate fitness competitions.

To perform whole genome sequencing, cells from mixed pools were spread onto M3 agarose plates, single colonies were selected and grown in YPD to saturation. DNA was extracted using the PureLink™ Genomic DNA Mini Kit (K182002). Sequencing libraries were made using Illumina DNA Prep kit by diluting reactions by 1/5. Briefly, samples were prepared such that the starting concentration in 6 µl was between 20 and 100 ng of DNA. 2 µl of BLT and TB1 were added to the starting material and incubated on a thermocycler at 55°C (lid 100°C) for 15 min. 2 µl of TSB was added to each reaction and incubated at 37°C (lid 100°C) for 15 min. Beads were washed 2 times with 20 µl of TWB. Following the final wash, 4 µl of EPM, 4 µl of water and 2 µl of UD indexes were added to each sample. Depending on starting concentration, PCR was performed based on Illumina guidelines as follows: lid 100°C, 68°C for 3 min, 98°C for 3 min, [98°C for 45s, 62°C for 30s, 68°C for 2min] for 6 to 10 cycles, 68°C for 1 min, 10°C hold. PCR products were cleaned with a double side sized selection as follows: 4 µl of each sample was pooled together (32 µl total for 8 samples) and added to 28 µl of water plus 32 µl of SPB. After a 5 min incubation 25 µl of supernatant was moved to a new tube containing 3 µl of SPB. Beads were washed with fresh 80% ethanol and libraries were eluted in 12 µl RSB. Samples were multiplexed using Illumina's unique dual (UD) index plates (A-D) and sequencing was performed with 2×150 paired end sequencing on HiSeq X at Psomagen (Rockville, MD).

In total 122 colonies were randomly picked and sequenced. As one might expect, barcodes that rose to high frequency were more likely to be picked multiple times. In an attempt to find lineages with unique attributes, some cultures were grown at 37°C or plated to high concentrations of drug prior to picking isolated colonies for sequencing. Of the 122 genomes we sequenced, only 53 had unique barcodes and pertained to the 774 lineages for which we obtained high enough barcode coverage to infer fitness. Only two of these 53 had no sequenced mutations suggesting their fitness increase over ancestor is due to a mutation we are unable to identify by sequencing, perhaps a change in ploidy. The other 51 all had at least one single nucleotide mutation in a gene reported in **supplemental table 1**.

Variant calling was done using GATK as described here: <https://github.com/gencorefacility/variant-calling-pipeline-gatk4> [↗](#). Identified variants were annotated using SnpEff (Cingolani et al. 2012 [↗](#)). Variant call files from 132 (53 unique/in CS) sequenced lineages were analyzed in R and compared to reference strain GCF\_000146045.2 (Genome assembly 64: sacCer3). SNPs present in the ancestor (as well as all evolved lineages) were ignored as these could not have caused the fitness differences we observed. We also ignored SNPs that were present in a substantial number of evolved lineages, as these likely represent background mutations that were present in a substantial portion of the cells representing the landing pad strain (SHA185). These are reported in **supplemental table 1** and include: SRD1-Glu97Lys, RSC30-Gly571Asp, OPT1-Val143Ile and LYS20-Thr29Met.

## Measuring growth curves of evolved lineages with unexpected fitness in multidrug conditions

Though fitness differences are not necessarily due to differences in maximum growth rate (Li et al. 2018b [↗](#)), we measured growth curves for a few lineages to investigate a case where an evolved lineage had unexpectedly low fitness in multidrug conditions (**Figure 7B** [↗](#)). Indeed, we found that this mutant grew more slowly in those conditions (**Figure 7D** [↗](#); **S10**). To perform this test, a lineage with a mutation to GBP2, a lineage with a mutation to HDA1, and sometimes the ancestor strain were streaked to YPD plates. We used the barcodeless ancestor strain, which is identical to the evolved lineages in every way except for lacking a barcode, and is described above in the methods section entitled, “Initiating barcoded fitness competition experiments”. A single colony of each strain was isolated from YPD plates and was used to inoculate an overnight YPD culture. After ~24 hours, a coulter counter (BD) was used to determine the number of cells/ml present in each culture. Next, all cultures were diluted such that the starting number of cells was 250,000 in 6 ml of M3 plus drug (either HR, LF, LR, LRLF or HRLF, see **Table 1** [↗](#)). To measure growth curves, these samples were allowed to grow at 30 degrees C. OD was measured every 10 minutes as the cultures were grown to saturation using the compact rocking incubator TVS062CA (Advantec Mfs). Raw growth curves for these conditions are shown in **Figure S10B and C**. Maximum growth rate was calculated using a sliding window approach to determine the region of each growth curve with the steepest log-linear slope.

## Determining ploidy

While our barcoded yeast strain is haploid, previous studies observed that some cells diploidize during the course of evolution in M3 media and by doing so gain a fitness advantage (Levy et al. 2015 [↗](#); Venkataram et al. 2016 [↗](#)). To ensure that observed fitness effects in our experiments were not largely due to the effects of diploids, we estimated the percent of diploid cells in each of our populations. We chose to make our estimates from frozen samples taken at the same time points from which we sampled 1 or 2 thousand cells to initiate fitness competitions (**Figure S1**). As such, our estimates also report on the percent of diploids that were present at the start of the fitness competitions experiments (**Table S2**).

To study ploidy, we used the nucleic acid stain SYTOX Green, which is capable of selectively staining the nucleus of fixed cells and has been shown to be optimal for use in budding yeast (Haase 2004 [↗](#)). For each of the 12 evolution experiments conditions, a small amount of freezer stock from the chosen timepoints (**Figure S1**) was plated to YPD and grown for ~48 hours. Individual colonies were picked and transferred to 96-well plates, one full plate per each condition, before being fixed with 95% ethanol for 1 hour. Plates were centrifuged at 4500 rpm and supernatant was discarded. 50 µL RNase A was added to the samples at a concentration of 2mg/mL, and the plates were then incubated for 2 hours at 37°C. Cells were pelleted by centrifuge and the supernatant was removed, which was followed by treatment with 20 µL of the protease pepsin at a concentration of 5mg/µL. Pepsin treated samples incubated at 37°C for 30 minutes before centrifugation and removal of supernatant. Finally, cells were resuspended in 50 µL TrisCL (50 mM, pH 8) and stained with 100 µL of 1µM SYTOX Green. Known diploid and haploid strains were used as controls alongside our samples to determine the expected fluorescence of stained diploid vs. haploid cells. Analysis was performed using a ThermoFisher Attune NxT, housed in the Flow Cytometry Core Facility at Arizona State University.

## Dimensional reduction

Our fitness inference procedure resulted in a data set consisting of nearly 10,000 fitness measurements (774 lineages x 12 conditions = 9288 fitness measurements). Dimensional reduction was performed on these data using UMAP (McInnes et al. 2018b [↗](#)). Clusters of similar mutants were identified and colored using a gaussian mixed model (Fraley and Raftery 2003 [↗](#)); Bayesian



Information Criteria were used to select the number of clusters (**Figure S6**). These analyses were performed in R; code can be found [https://osf.io/pxyv9/?view\\_only=51241d8f00c24f7e83f1ece3ae31a53b](https://osf.io/pxyv9/?view_only=51241d8f00c24f7e83f1ece3ae31a53b).

In order to prevent conditions with the most variation in fitness (e.g., high FLU) from dominating, we normalized fitness measurements from each of the 12 environments to have the same overall mean and variance (we transformed the data from every environment to have a mean of 0 and a standard deviation of 1) before performing dimensional reduction. This normalization procedure did not have a dramatic effect on the UMAP (**Figure S7A**). We also explored normalizing all data to account for magnitude differences by setting the average fitness of each lineage across all 12 environments to 0. Doing so did not significantly change the groupings present in the UMAP from those displayed in **Figure 4** (**Figure S9**) other than in the ways we describe in **Figure 6**. Reducing our data set to 617 adaptive lineages with very high sequencing coverage (**Figure S7B**) also did not significantly affect the way that mutants cluster into groups, nor did using a different dimensional reduction algorithm altogether (**Figure S8** and see next paragraph). In short, the clustering of mutants was robust to the different decisions we made when choosing how to analyze these data.

In order to assess whether clusters identified from the UMAP are robust to alternative clustering methods, we also used hierarchical clustering to identify clusters of mutants with similar fitness profiles. First, we computed the pairwise distance of all lineages across the fitness profiles. Then, we used Ward's method from scikit-learn to iteratively cluster lineages such that the within-cluster variation is minimized (Ward 1963; Pedregosa et al. 2012). To test the consistency of lineage clustering, we chose a pairwise cluster distance cutoff of 11, which results in the same number of clusters (7) as identified with the UMAP clustering approach used in the main text. We then compared the identity of the lineages within each of these clusters with the UMAP clusters. We found that, for most clusters, over 80% of lineages from the UMAP cluster corresponded with a unique hierarchical cluster and labeled these hierarchical clusters according to this correspondence (**Figure S8**). For UMAP cluster 1, lineages were more evenly split between two clusters. 64% of these lineages clustered together in what is labeled as hierarchical cluster 1 and 30% in hierarchical cluster 1/7 (**Figure S8**), which contains all of the control lineages that comprise UMAP cluster 7. Despite these lineages clustering more closely with control lineages than the remainder of the cluster 1, they do tend to cluster distinctly with the control lineages, suggesting they have behavior that is distinguishable from the control lineages. If we consider these cluster 1 mutants that end up in cluster 3/7 as “mis-clustered”, we find that 85% of lineages from each UMAP cluster are clustered together in the corresponding hierarchical cluster. If we consider these as “consistently clustered”, this metric increases to 90% of lineages correctly clustered. Altogether, this analysis shows that the results we show are robust to alternative methods of clustering.

## Inferring where adaptive lineages originally evolved

All 774 adaptive lineages were isolated from one of the 12 evolution experiments at the timepoint indicated in **figure S1** (see Methods section entitled, “**Isolating a large pool of adaptive mutants**”). The sample we isolated from each evolution experiment was sequenced prior to pooling. This allows us to computationally determine which barcoded lineages originated from which evolution experiment to generate the pie charts in **Figures 3**, **5**, **6**, **7** and **8**.

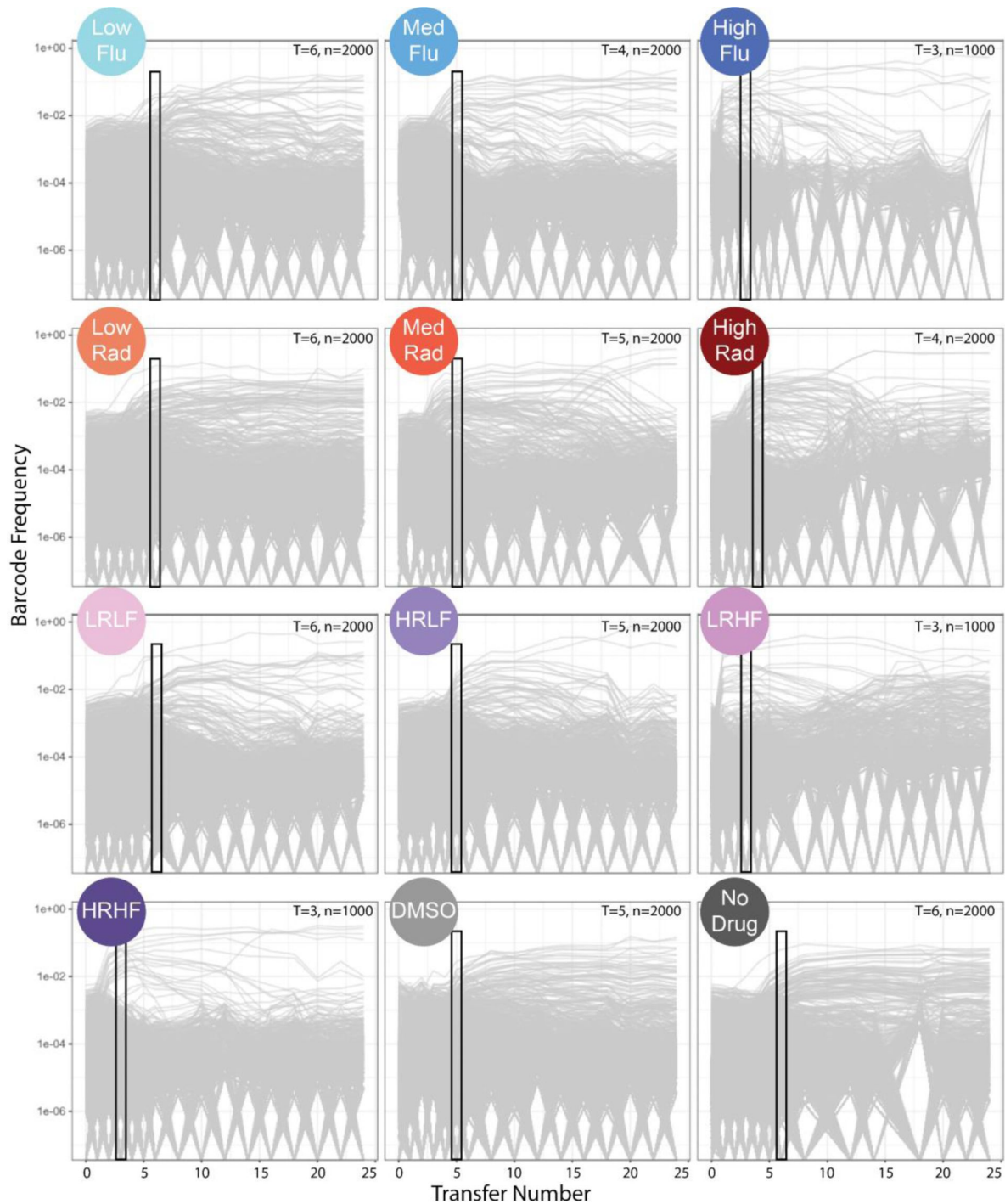
If adaptive mutation arose independently during the course of each evolution experiment, it would be unlikely for any adaptive lineage we study to be present in more than one of the evolution conditions. This would make it very easy to assign each barcode to the evolution experiment from which it originated. However, this was not the case for many barcoded lineages.

Previous work explained that the transformation procedure used to insert a barcode into the landing pad of SHA185 is itself mutagenic, such that many mutations arise prior to the start of the evolution experiments (Levy et al. 2015). Since all our evolution experiments were started from



the same pool of barcoded lineages, we thus expect that many adaptive lineages will be present in more than one condition. However, it is not expected that these adaptive lineages will be present at the same frequency in every condition; instead these frequencies change with the fitness of the mutation each lineage possesses. Therefore, when an adaptive lineage appeared in multiple conditions, we weighted its origin to reflect its frequency in each condition. In other words, adaptive lineages that were only present in the sample taken from a single evolution condition were identified and assigned a single origin condition in the pie charts in **Figures 3**, **5**, **6**, **7**, and **8**. But for adaptive lineages found in the samples taken from more than one evolution condition, the proportions assigned to each origin condition in the pie charts was scaled to equal the relative frequencies of that lineage in all evolution conditions where it was observed. Associated data and code can be found here: [https://osf.io/pxyv9/?view\\_only=51241d8f00c24f7e83f1ece3ae31a53b](https://osf.io/pxyv9/?view_only=51241d8f00c24f7e83f1ece3ae31a53b).

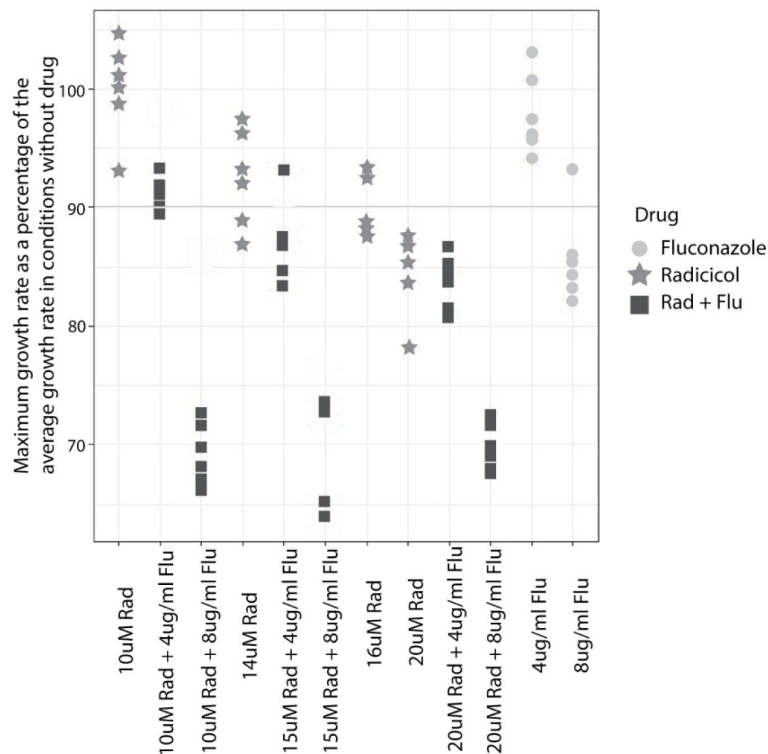
## Supplemental Figures and Tables



**Figure S1**

**Twelve barcoded evolution experiments track ~300,000 lineages as they adapt to different drug concentrations and combinations.**

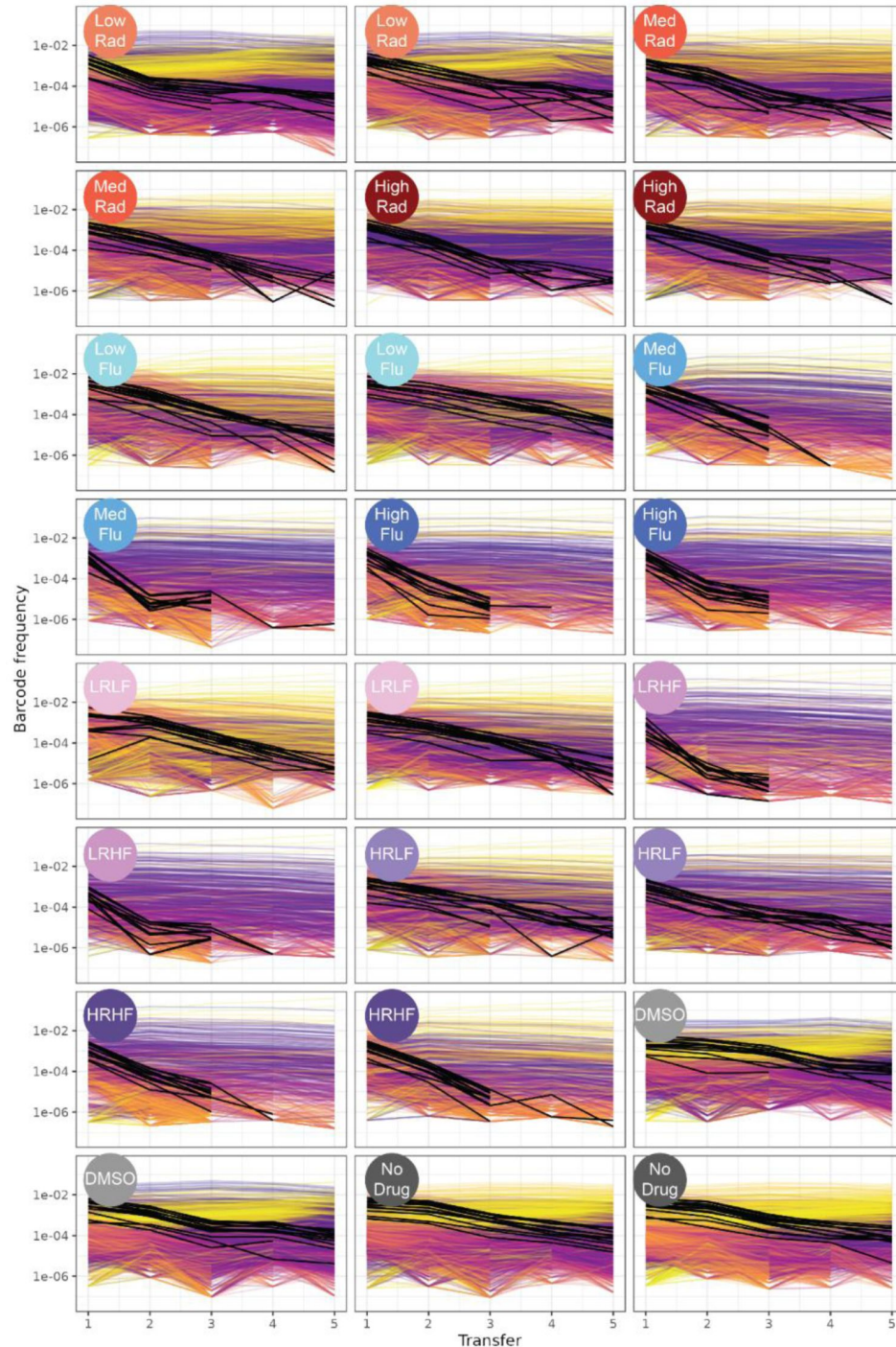
In each plot, every line represents a unique barcoded lineage. The vertical axis represents barcode frequency. When lineages reach a frequency of zero it means they were not sampled at that time point of the experiment. The black boxes indicate the transfer number of each evolution experiment from which evolved lineages are sampled; the number of cells sampled (colonies picked) is in the upper right hand corner.



**Figure S2**

**Chosen drug concentrations do not dramatically reduce yeast's maximum growth rate.**

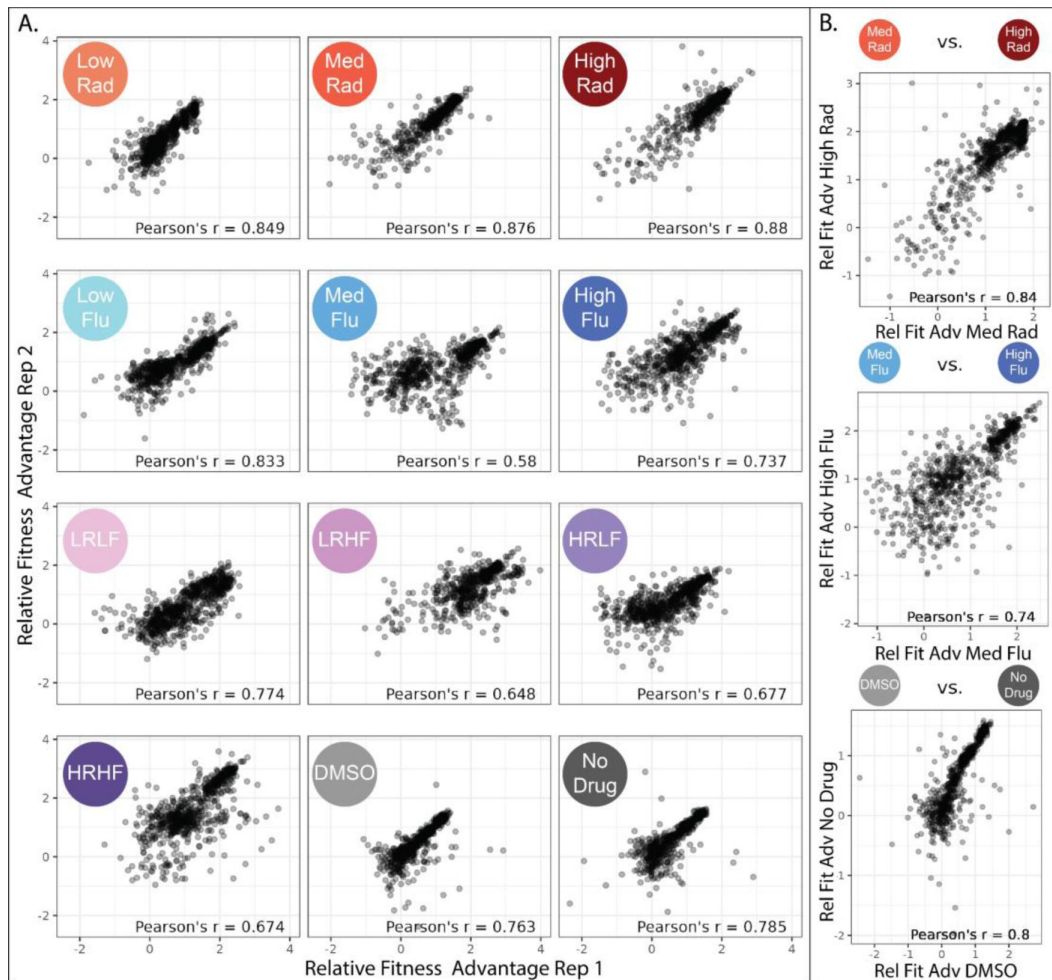
We measured the growth rate of a single barcoded yeast lineage, prior to the evolution experiment, in different concentrations and combinations of drugs using a plate reader to track changes in optical density over time. Maximum growth rate is reported as a percentage of the maximum growth rate in conditions lacking FLU or RAD. Maximum growth rate was calculated as the maximum log-linear slope of the change in optical density over time.



**Figure S3**

**Twenty four fitness competitions track evolved lineages as their barcodes change frequency.**

In each plot, every line represents a barcoded lineage. Lineages with >5 reads per experiment are shown (4,815 lineages shown). Control lineages, known to possess no fitness differences relative to the ancestor, are highlighted in black. Evolved lineages with higher fitness than controls are yellow, similar fitness to controls (i.e., relative fitness ~0) are purple, and lower fitness than controls are orange. When a lineage declines in frequency so much that its associated barcode is no longer observed, its line abruptly ends. Some lineages appear to decline and then increase in frequency; this happens because low frequency lineages are subject to sampling noise.

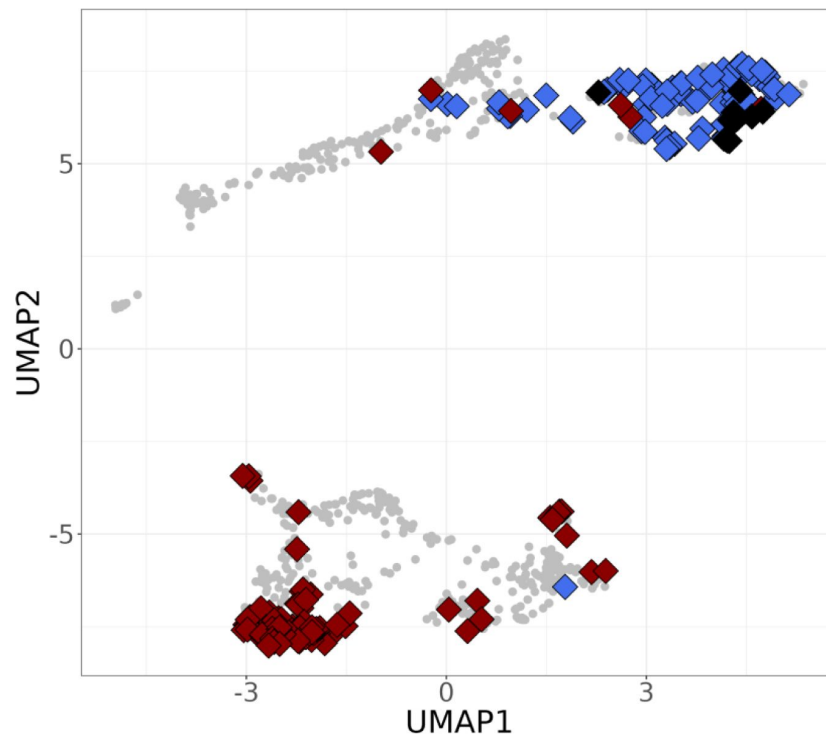


**Figure S4**

**Fitness measurements are reproducible between replicates and closely related conditions.**

**A.** Every point depicts one of the evolved lineages that was observed more than 500 times in both replicates of the fitness competition experiment for a given condition. The average Pearson correlation across all pairs of replicates in all conditions is 0.75. **B.** Every point depicts one of the 774 lineages that was observed more than 500 times in at least one replicate fitness competition. Here, instead of comparing the fitness of replicates, we compare fitness across conditions for 3 closely related pairs of conditions. Comparisons across conditions may be less noisy than those across replicates because fitness in a particular condition represents the average fitness across replicates in that condition.

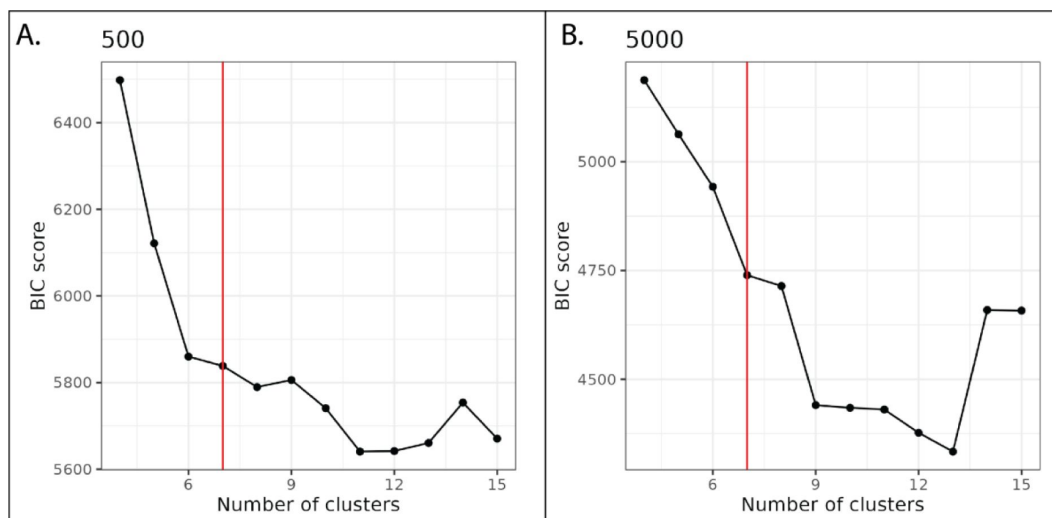




**Figure S5**

**The two types of adaptive mutants depicted in figure 3 sort into different clusters on the UMAP.**

The top 100 highest fitness lineages in high FLU (blue) and high RAD (red) largely cluster into different groups, with the 7 overlapping mutants (black) falling into the uppermost group.

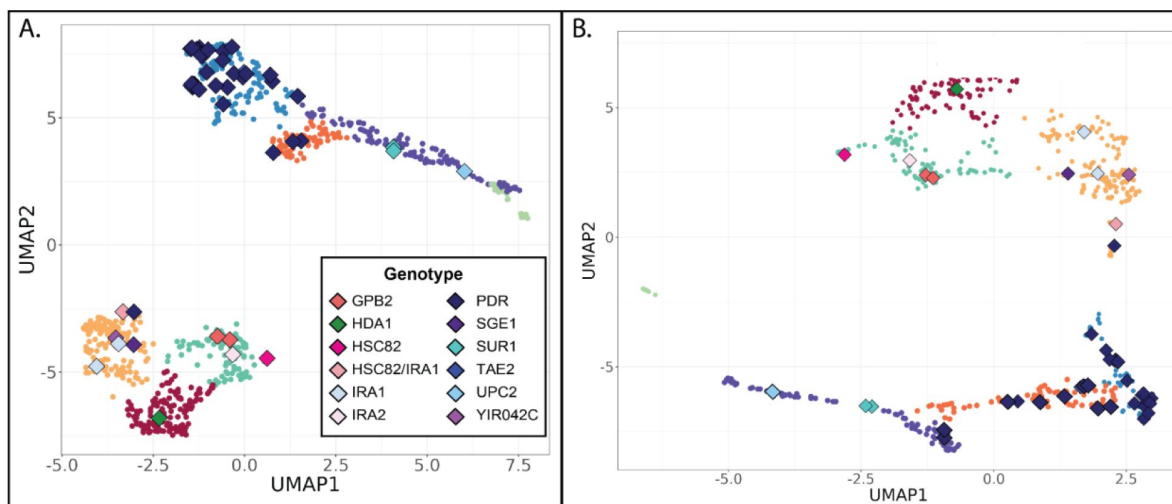


**Figure S6**

**Bayesian information criteria (BIC) scores suggest the 774 mutants cluster into between 6 and 13 groups.**

We used a gaussian mixture model to distinguish clusters of mutants with unique fitness profiles. **A.** BIC scores for analysis performed on 774 mutant lineages that each were observed at minimum 500 times per environment. **B** BIC scores for analysis performed on 617 mutant lineages that each were observed at minimum 5000 times per environment.

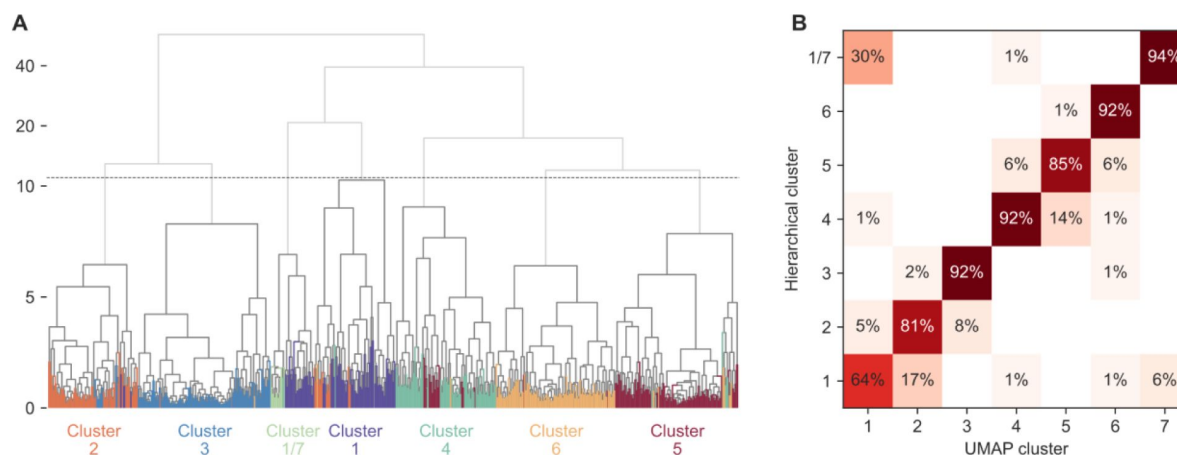




**Figure S7.**

### UMAP structure is robust.

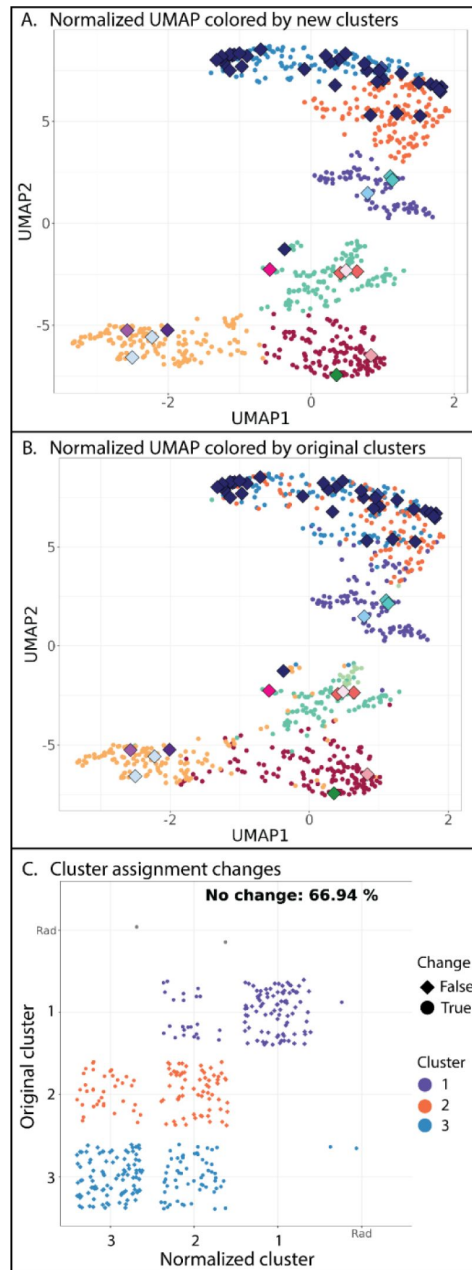
**A.** In this UMAP plot, relative fitness of the 774 lineages is fed directly into UMAP prior to normalizing the data to have equal variance across all environments as was done in [figure 4D](#). This has little impact on the appearance of the UMAP. **B.** UMAP made with 617 lineages that were observed more than 5000 times in at least one replicate. While the appearance of the UMAP is inverted relative to [figure 4D](#), the clusters largely retain the same lineages. As lower coverage (noisier) data points are removed from this dataset, the observation that the overall patterns are maintained suggests our original clusters were not based on noisy fitness measurements.



**Figure S8.**

### Clusters are robust to a hierarchical clustering method.

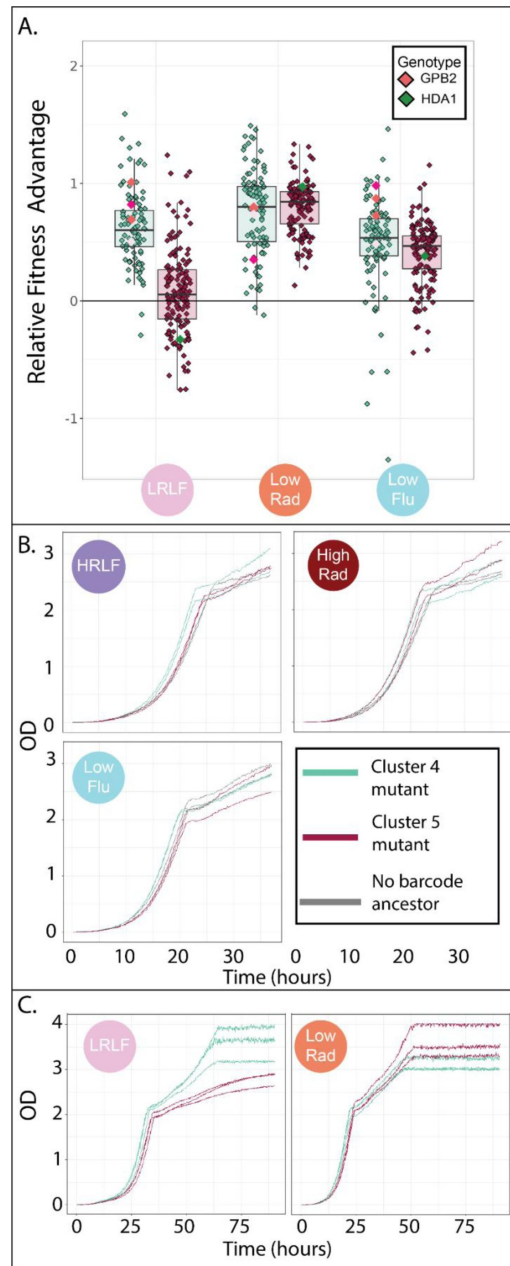
**A.** In order to assess whether clusters identified from the UMAP are robust to alternative clustering methods, we used hierarchical clustering to identify clusters of mutants with similar fitness profiles. This figure depicts a dendrogram where the branch length between each lineage (or group of lineages) represents the distance between the clustered groups as measured across all measured conditions. The terminal branch leading to each lineage is colored by that lineage's cluster from the UMAP clustering used in [figure 4](#) of the main text. Dark gray lines show clusters identified using a between-cluster cutoff of 11, which creates 7 clusters, consistent with the number of clusters we identified in [figure 4](#) via clustering on the UMAP space. Hierarchical clusters are labeled and colored corresponding to whichever of the UMAP clusters they share the most overlapping lineages with. Note that the y-axis is linearly scaled between 0 and 10 and log2 scaled above 10 for visualization purposes. **B.** A matrix quantifying how well the results of this hierarchical clustering method correspond with those of the UMAP method presented in [figure 4](#), depicting the percentage of a given UMAP cluster that ends up with each corresponding hierarchical cluster. For the most part, mutants that are clustered together in the UMAP in [figure 4](#) are also clustered together in [panel A](#). Even though it appears that mutants in cluster 1 (which are those most resistant to low FLU) move to a different cluster, these mutants move together to the cluster pertaining to the control strains as they have the least severe effects on fitness.



**Figure S9**

### UMAP on data that were normalized to account for magnitude differences (row means set to 0).

**A.** Every point in this plot represents one of the 774 barcoded lineages colored by cluster; clusters were identified using a gaussian mixture model. After normalizing to account for magnitude differences, the 774 adaptive lineages cluster into 6 (not 7) groups based on variation in their fitness profiles. The number of clusters is reduced relative to the UMAP in [figure 4](#) because the control lineages no longer form a separate group; they cluster into the middle cluster in light green. **B.** This plot shows the same UMAP as above in [panel A](#), but the color of each of the 774 points is changed to reflect its color in the UMAP in [figure 4](#). Despite coloring based on a different UMAP, differently colored points still largely fall into separate clusters, except for those points colored blue or orange, which are discussed in [figure 6](#). **C.** This plot depicts all lineages in clusters 1 – 3 in either the [figure 4](#) UMAP (not normalized to lessen magnitude differences) or the above UMAP (normalized to lessen magnitude differences). Lineages that do not switch from one cluster to another are represented by diamonds while those that switch are noted with circles. Across these three clusters, 67% of lineages remain in their original cluster.



**Figure S10**

### Unexpected tradeoffs in evolved lineages in cluster 4 and 5 in response to combined drugs.

**A.** Cluster 5 (red) has an unexpected fitness disadvantage in the LRLF multidrug environment relative to cluster 4 (green), given that cluster 5 lineages do not have a relative fitness disadvantage in the relevant single drug environments. This plot is similar to the one in [figure 7B](#) except here the multidrug condition shown is LRLF rather than HRLF. Boxplots summarize the distribution across all lineages within each cluster in each environment, displaying the median (center line), interquartile range (IQR) (upper and lower hinges), and highest value within  $1.5 \times \text{IQR}$  (whiskers). **B.** Raw growth curve data from which relative maximum growth rates presented in [Figure 7D](#) were calculated. Two mutants, one to GBP2 (cluster 4; green) and one to HDA1 (cluster 5; red) were each grown individually in the conditions shown, as was the barcodeless ancestor strain (grey). All experiments were conducted in duplicate. We initiated all cultures from the same number of starting cells. The mutant in cluster 5 (red) has an obvious growth disadvantage in the combination condition (HRLF). **C.** Raw growth curve data for the same two mutants in [panel B](#) in different conditions. These growth experiments were conducted similarly to those in [panel B](#), except they were tracked for a longer period of time, lacked the ancestor strain, and involved 3 replicates per mutant. There appear to be obvious and reproducible differences in the growth curves of the two mutants surveyed.

| Evolution Conditions | Diploids detected (96 sampled) | Estimated % diploid |
|----------------------|--------------------------------|---------------------|
| Low Rad              | 8                              | 8.3%                |
| Med Rad              | 0                              | 0.0%                |
| High Rad             | 0                              | 0.0%                |
| Low Flu              | 9                              | 9.4%                |
| Med Flu              | 0                              | 0.0%                |
| High Flu             | 1                              | 1.0%                |
| LR + LF              | 4                              | 4.2%                |
| HR +LF*              | 1                              | 1.1%                |
| LR + HF              | 1                              | 1.0%                |
| HR + HF              | 3                              | 3.1%                |
| DMSO                 | 6                              | 6.3%                |
| No Drug              | 9                              | 9.4%                |

**Table S2**

**Estimated percentage of diploid cells in each evolution condition at the time point we sampled, determined using nuclear staining and flow cytometry as described in Materials and Methods. \*For the High Radicicol + Low Fluconazole condition, 95 instead of 96 samples were measured due to undetectable growth in one well.**



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## Editors

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**Christian Landry**

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### Reviewer #1 (Public Review):

#### Summary:

In their manuscript, Schmidlin, Apodaca, et al try to answer fundamental questions about the evolution of new phenotypes and the trade-offs associated with this process. As a model, they use yeast resistance to two drugs, fluconazole and radicicol. They use barcoded libraries of isogenic yeasts to evolve thousands of strains in 12 different environments. They then measure the fitness of evolved strains in all environments and use these measurements to examine patterns in fitness trade-offs. They identify only six major clusters corresponding to different trade-off profiles, suggesting the vast genotypic landscape of evolved mutants translates to a highly constrained phenotypic space. They sequence over a hundred evolved strains and find that mutations in the same gene can result in different phenotypic profiles.

Overall, the authors deploy innovative methods to scale up experimental evolution experiments, and in many aspects of their approach tried to minimize experimental variation.

#### Weaknesses:

(1) One of the objectives of the authors is to characterize the extent of phenotypic diversity in terms of resistance trade-offs between fluconazole and radicicol. To minimize noise in the measurement of relative fitness, the authors only included strains with at least 500 barcode counts across all time points in all 12 experimental conditions, resulting in a set of 774 lineages passing this threshold. This corresponds to a very small fraction of the starting set of ~21 000 lineages that were combined after experimental evolution for fitness measurements. As the authors briefly remark, this will bias their datasets for lineages with high fitness in all 12 environments, as all these strains must be fit enough to maintain a high abundance. One of the main observations of the authors is phenotypic space is constrained to a few clusters of roughly similar relative fitness patterns, giving hope that such clusters could be enumerated and considered to design antimicrobial treatment strategies. However, by excluding all lineages that fit in only one or a few environments, they conceal much of the diversity that might exist in terms of trade-offs and set up an inclusion threshold that might present only a small fraction of phenotypic space with characteristics consistent with generalist resistance mechanisms or broadly increased fitness. This has important implications regarding the general conclusions of the authors regarding the evolution of trade-offs.

(2) Most large-scale pooled competition assays using barcodes are usually stopped after ~25 to avoid noise due to the emergence of secondary mutations. The authors measure fitness across ~40 generations, which is almost the same number of generations as in the evolution experiment. This raises the possibility of secondary mutations biasing abundance values, which would not have been detected by the whole genome sequencing as it was performed before the competition assay.

(3) The approach used by the authors to identify and visualize clusters of phenotypes among lineages does not seem to consider the uncertainty in the measurement of their relative fitness. As can be seen from Figure S4, the inter-replicate difference in measured fitness can often be quite large. From these graphs, it is also possible to see that some of the fitness

measurements do not correlate linearly (ex.: Med Flu, Hi Rad Low Flu), meaning that taking the average of both replicates might not be the best approach. Because the clustering approach used does not seem to take this variability into account, it becomes difficult to evaluate the strength of the clustering, especially because the UMAP projection does not include any representation of uncertainty around the position of lineages. This might paint a misleading picture where clusters appear well separate and well defined but are in fact much fuzzier, which would impact the conclusion that the phenotypic space is constricted.

(4) The authors make the decision to use UMAP and a gaussian mixed model to cluster and represent the different fitness landscapes of their lineages of interest. Their approach has many caveats. First, compared to PCA, the axis does not provide any information about the actual dissimilarities between clusters. Using PCA would have allowed a better understanding of the amount of variance explained by components that separate clusters, as well as more interpretable components. Second, the advantages of dimensional reduction are not clear. In the competition experiment, 11/12 conditions (all but the no drug, no DMSO conditions) can be mapped to only three dimensions: concentration of fluconazole, concentration of radicicol, and relative fitness. Each lineage would have its own fitness landscape as defined by the plane formed by relative fitness values in this space, which can then be examined and compared between lineages. Third, the choice of 7 clusters as the cutoff for the multiple Gaussian model is not well explained. Based on Figure S6A, BIC starts leveling off at 6 clusters, not 7, and going to 8 clusters would provide the same reduction as going from 6 to 7. This choice also appears arbitrary in Figure S6B, where BIC levels off at 9 clusters when only highly abundant lineages are considered. This directly contradicts the statement in the main text that clusters are robust to noise, as more a stringent inclusion threshold appears to increase and not decrease the optimal number of clusters. Additional criteria to BIC could have been used to help choose the optimal number of clusters or even if mixed Gaussian modeling is appropriate for this dataset.

(5) Large-scale barcode sequencing assays can often be noisy and are generally validated using growth curves or competition assays. Having these types of results would help support the accuracy of the main assay in the manuscript and thus better support the claims of the authors.

<https://doi.org/10.7554/eLife.94144.1.sa1>

#### **Reviewer #2 (Public Review):**

##### **Summary:**

Schmidlin & Apodaca et al. aim to distinguish mutants that resist drugs via different mechanisms by examining fitness tradeoffs across hundreds of fluconazole-resistant yeast strains. They barcoded a collection of fluconazole-resistant isolates and evolved them in different environments with a view to having relevance for evolutionary theory, medicine, and genotype-phenotype mapping.

##### **Strengths:**

There are multiple strengths to this paper, the first of which is pointing out how much work has gone into it; the quality of the experiments (the thought process, the data, the figures) is excellent. Here, the authors seek to induce mutations in multiple environments, which is a really large-scale task. I particularly like the attention paid to isolates with are resistant to low concentrations of FLU. So often these are overlooked in favour of those conferring MIC values >64/128 etc. What was seen is different genotype and fitness profiles. I think there's a wealth of information here that will actually be of interest to more than just the fields mentioned (evolutionary medicine/theory).

#### Weaknesses:

Not picking up low fitness lineages - which the authors discuss and provide a rationale as to why. I can completely see how this has occurred during this research, and whilst it is a shame I do not think this takes away from the findings of this paper. Maybe in the next one!

In the abstract the authors focus on 'tradeoffs' yet in the discussion they say the purpose of the study is to see how many different mechanisms of FLU resistance may exist (lines 679-680), followed up by "We distinguish mutants that likely act via different mechanisms by identifying those with different fitness tradeoffs across 12 environments". Whilst I do see their point, and this is entirely feasible, I would like a bit more explanation around this (perhaps in the intro) to help lay-readers make this jump. The remainder of my comments on 'weaknesses' are relatively fixable, I think:

In the introduction I struggle to see how this body of research fits in with the current literature, as the literature cited is a hodge-podge of bacterial and fungal evolution studies, which are very different! So example, the authors state "previous work suggests that mutants with different fitness tradeoffs may affect fitness through different molecular mechanisms" (lines 129-131) and then cite three papers, only one of which is a fungal research output. However, the next sentence focuses solely on literature from fungal research. Citing bacterial work as a foundation is fine, but as you're using yeast for this I think tailoring the introduction more to what is and isn't known in fungi would be more appropriate. It would also be great to then circle back around and mention monotherapy vs combination drug therapy for fungal infections as a rationale for this study. The study seems to be focused on FLU-resistant mutants, which is the first-line drug of choice, but many (yeast) infections have acquired resistance to this and combination therapy is the norm.

Methods: Line 769 - which yeast? I haven't even seen mention of which species is being used in this study; different yeast employ different mechanisms of adaptation for resistance, so could greatly impact the results seen. This could help with some background context if the species is mentioned (although I assume *S. cerevisiae*). In which case, should aneuploidy be considered as a mechanism? This is mentioned briefly on line 556, but with all the sequencing data acquired this could be checked quickly?

I think the authors could be bolder and try and link this to other (pathogenic) yeasts. What are the implications of this work on say, *Candida* infections?

<https://doi.org/10.7554/eLife.94144.1.sa0>

#### Author Response

We thank both reviewers for taking the time and effort to think critically about our paper and point out areas where it can be improved. In this document, we do our best to clarify any misunderstandings with the hope that further consideration about the strengths and weaknesses of our approach will be possible. Our responses are in bold.

##### **Reviewer #1 (Public Review):**

##### *Summary:*

*In their manuscript, Schmidlin, Apodaca, et al try to answer fundamental questions about the evolution of new phenotypes and the trade-offs associated with this process. As a model, they use yeast resistance to two drugs, fluconazole and radicicol. They use*



## Public Reviews

We thank both reviewers for taking the time and effort to think critically about our paper and point out areas where it can be improved. In this document, we do our best to clarify any misunderstandings with the hope that further consideration about the strengths and weaknesses of our approach will be possible. Our responses are in bold.

### Reviewer #1 (Public Review):

#### Summary:

*In their manuscript, Schmidlin, Apodaca, et al try to answer fundamental questions about the evolution of new phenotypes and the trade-offs associated with this process. As a model, they use yeast resistance to two drugs, fluconazole and radicicol. They use barcoded libraries of isogenic yeasts to evolve thousands of strains in 12 different environments. They then measure the fitness of evolved strains in all environments and use these measurements to examine patterns in fitness trade-offs. They identify only six major clusters corresponding to different trade-off profiles, suggesting the vast genotypic landscape of evolved mutants translates to a highly constrained phenotypic space. They sequence over a hundred evolved strains and find that mutations in the same gene can result in different phenotypic profiles.*

*Overall, the authors deploy innovative methods to scale up experimental evolution experiments, and in many aspects of their approach tried to minimize experimental variation.*

We thank the reviewer for this positive assessment of our work. We are happy that the reviewer noted what we feel is a unique strength of our approach: we scaled up experimental evolution by using DNA barcodes and by exploring 12 related selection pressures. Despite this scaling up, we still see phenotypic convergence among the 744 adaptive mutants we study.

The environments we study represent 12 different concentrations or combinations of two drugs, radicicol and fluconazole. Our hope is that this large dataset (774 mutants x 12 environments) will be useful, both to scientists who are generally interested in the genetic and phenotypic underpinnings of adaptation, and to scientists specifically interested in the evolution of drug resistance.

#### Weaknesses:

*(1) One of the objectives of the authors is to characterize the extent of phenotypic diversity in terms of resistance trade-offs between fluconazole and radicicol. To minimize noise in the measurement of relative fitness, the authors only included strains with at least 500 barcode counts across all time points in all 12 experimental conditions, resulting in a set of 774 lineages passing this threshold. This corresponds to a very small fraction of the starting set of ~21 000 lineages that were combined after experimental evolution for fitness measurements.*

This is a misunderstanding that we will work to clarify in the revision. Our starting set did not include 21,000 adaptive lineages. The total number of unique adaptive lineages in this starting set is much lower than 21,000 for two reasons.

First, ~21,000 represents the number of single colonies we isolated in total from our evolution experiments. Many of these isolates possess the same barcode, meaning they are duplicates. Second, and more importantly, most evolved lineages do not acquire adaptive mutations, meaning that many of the 21,000 isolates are genetically identical to their ancestor. In our

revised manuscript, we will explicitly state that these 21,000 isolated lineages do not all represent unique, adaptive lineages. In figure 2 and all associated text, we will change the word “lineages” to “isolates,” where relevant.

More broadly speaking, several previous studies have demonstrated that diverse genetic mutations converge at the level of phenotype, and have suggested that this convergence makes adaptation more predictable (PMID33263280, PMID37437111, PMID22282810, PMID25806684). Our study captures mutants that are overlooked in previous studies, such as those that emerge across subtly different selection pressures (e.g., 4  $\mu$ g/ml vs. 8  $\mu$ g/ml flu) and those that are undetectable in evolutions lacking DNA barcodes. Thus, while our experimental design misses some mutants (see next comment), it captures many others. Note that 774 adaptive lineages is more than most previous studies. Thus, we feel that “our work – showing that 774 mutants fall into a much smaller number of groups” is important because it “contributes to growing literature suggesting that the phenotypic basis of adaptation is not as diverse as the genetic basis (lines 161 - 162).”

*As the authors briefly remark, this will bias their datasets for lineages with high fitness in all 12 environments, as all these strains must be fit enough to maintain a high abundance.*

The word “briefly” feels a bit unfair because we discuss this bias on 3 separate occasions (on lines 146 - 147, 260 - 264, and in more detail on 706 - 714). We even walk through an example of a class of mutants that our study misses. We say, “our study is underpowered to detect adaptive lineages that have low fitness in any of the 12 environments. This is bound to exclude large numbers of adaptive mutants. For example, previous work has shown some FLU resistant mutants have strong tradeoffs in RAD (Cowen and Lindquist 2005). Perhaps we are unable to detect these mutants because their barcodes are at too low a frequency in RAD environments, thus they are excluded from our collection of 774.”

In our revised version, we will add more text to the first mention of these missing mutants (lines 146 - 147) so that the implications are more immediately made apparent.

While we “miss” some classes of mutants, we “catch” other classes that may have been missed in previous studies of convergence. For example, we observe a unique class of FLU-resistant mutants that primarily emerged in evolution experiments that lack FLU (Figure 3). Thus, we think that the unique design of our study, surveying 12 environments, allows us to make a novel contribution to the study of phenotypic convergence.

*One of the main observations of the authors is phenotypic space is constrained to a few clusters of roughly similar relative fitness patterns, giving hope that such clusters could be enumerated and considered to design antimicrobial treatment strategies. However, by excluding all lineages that fit in only one or a few environments, they conceal much of the diversity that might exist in terms of trade-offs and set up an inclusion threshold that might present only a small fraction of phenotypic space with characteristics consistent with generalist resistance mechanisms or broadly increased fitness. This has important implications regarding the general conclusions of the authors regarding the evolution of trade-offs.*

We discussed these implications in some detail in the 16 lines mentioned above (146 - 147, 260 - 264, 706 - 714). To add to this discussion, we will also add the following sentence to the end of the paragraph on lines 697 - 714: “This could complicate (or even make impossible) endeavors to design antimicrobial treatment strategies that thwart resistance”.

We will also add a new paragraph that discusses these implications earlier in our manuscript. This paragraph will highlight the strengths of our method (e.g., that we “catch” classes of

mutants that are often overlooked) while being transparent about the weaknesses of our approach (e.g., that we “miss” mutants with strong tradeoffs).

*(2) Most large-scale pooled competition assays using barcodes are usually stopped after ~25 to avoid noise due to the emergence of secondary mutations.*

The rate at which new mutations enter a population is driven by various factors such as the mutation rate and population size, so choosing an arbitrary threshold like 25 generations is difficult.

We conducted our fitness competition following previous work using the Levy/Blundell yeast barcode system, in which the number of generations reported varies from 32 to 40 (PMID33263280, PMID27594428, PMID37861305, see PMID27594428 for detailed calculation of the fraction of lineages biased by secondary mutations in this system).

*The authors measure fitness across ~40 generations, which is almost the same number of generations as in the evolution experiment. This raises the possibility of secondary mutations biasing abundance values, which would not have been detected by the whole genome sequencing as it was performed before the competition assay.*

We understand how the reviewer came to this misunderstanding and will adjust our revised manuscript accordingly. Previous work has demonstrated that, in this particular evolution platform, most of the mutations actually occur during the transformation that introduces the DNA barcodes (PMID25731169). In other words, these mutations do not accumulate during the 40 generations of evolution, they are already there. So the observation that we collect a genetically diverse pool of adaptive mutants after 40 generations of evolution is not evidence that 40 generations is enough time for secondary mutations to bias abundance values.

*(3) The approach used by the authors to identify and visualize clusters of phenotypes among lineages does not seem to consider the uncertainty in the measurement of their relative fitness. As can be seen from Figure S4, the inter-replicate difference in measured fitness can often be quite large. From these graphs, it is also possible to see that some of the fitness measurements do not correlate linearly (ex.: Med Flu, Hi Rad Low Flu), meaning that taking the average of both replicates might not be the best approach.*

This concern, and all subsequent concerns, seem to be driven by either (a) general concerns about the noisiness of fitness measurements obtained from large-scale barcode fitness assays or (b) general concerns about whether the clusters obtained from our dimensional reduction approach capture this noise as opposed to biologically meaningful differences.

We will respond to each concern point-by-point, but want to start by generally stating that (a) our particular large-scale barcode fitness assay has several features that diminish noise, and (b) we devote 4 figures and 200 lines of text to demonstrating that these clusters capture biologically meaningful differences between mutants (and not noise).

In terms of this specific concern, we performed an analysis of noise in the submitted manuscript: Our noisiest fitness measurements correspond to barcodes that are the least abundant and thus suffer the most from stochastic sampling noise. These are also the barcodes that introduce the nonlinearity the reviewer mentions. We removed these from our dataset by increasing our coverage threshold from 500 reads to 5,000 reads. The clusters did not collapse, which suggests that they were not capturing noise (Figure S7 panel B). But we agree with the reviewer that this analysis alone is not sufficient to conclude that the clusters distinguish groups of mutants with unique fitness tradeoffs.

*Because the clustering approach used does not seem to take this variability into account, it becomes difficult to evaluate the strength of the clustering, especially because the UMAP projection does not include any representation of uncertainty around the position of lineages.*

To evaluate the strength of the clustering, we performed numerous analyses including whole genome sequencing, growth experiments, reclustering, and tracing the evolutionary origins of each cluster (Figures 5 - 8). All of these analyses suggested that our clusters capture groups of mutants that have different fitness tradeoffs. We will adjust our revised manuscript to make clear that we do not rely on the results of a clustering algorithm alone to draw conclusions about phenotypic convergence.

We are also grateful to the reviewer for helping us realize that, as written, our manuscript is not clear with regard to how we perform clustering. We are not using UMAP to decide which mutant belongs to which cluster. Recent work highlights the importance of using an independent clustering method (PMID37590228). Although this recent work addresses the challenge of clustering much higher dimensional data than we survey here, we did indeed use an independent clustering method (gaussian mixture model). In other words, we use UMAP for visualization but not clustering. We also confirm our clustering results using a second independent method (hierarchical clustering; Figure S8). And in our revised manuscript, will confirm with a third method (PCA, see below). We will adjust the main text and the methods section to make these choices clearer.

*This might paint a misleading picture where clusters appear well separate and well defined but are in fact much fuzzier, which would impact the conclusion that the phenotypic space is constricted.*

The salient question is whether the clusters are so “fuzzy” that they are not meaningful. That interpretation seems unreasonable. Our clusters group mutants with similar genotypes, evolutionary histories, and fitness tradeoffs (Figures 5 - 8). Clustering mutants with similar behaviors is important and useful. It improves phenotypic prediction by revealing which mutants are likely to have at least some phenotypic effects in common. And it also suggests that the phenotypic space is constrained, at least to some degree, which previous work suggests is helpful in predicting evolution (PMID33263280, PMID37437111, PMID22282810, PMID25806684).

*(4) The authors make the decision to use UMAP and a gaussian mixed model to cluster and represent the different fitness landscapes of their lineages of interest. Their approach has many caveats. First, compared to PCA, the axis does not provide any information about the actual dissimilarities between clusters. Using PCA would have allowed a better understanding of the amount of variance explained by components that separate clusters, as well as more interpretable components.*

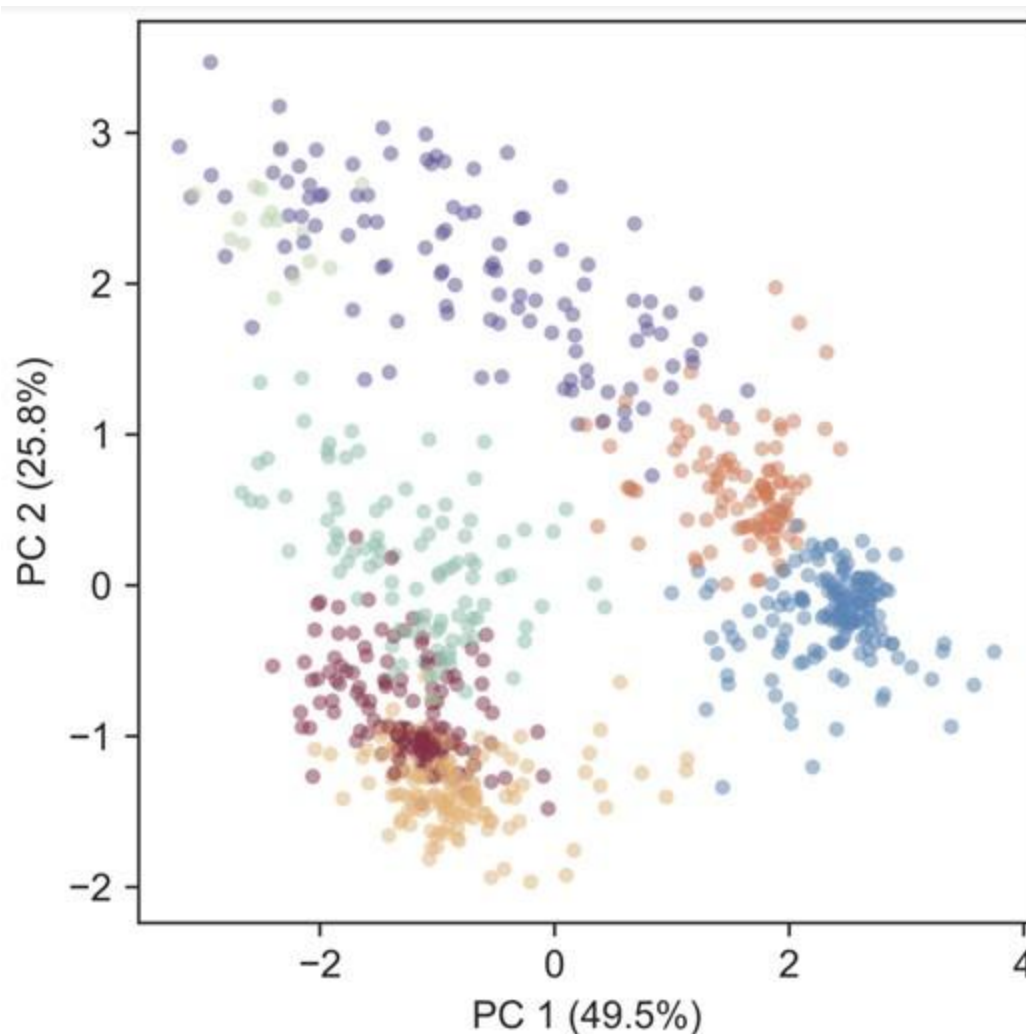
The components derived from PCA are often not interpretable. It's not obvious that each one, or even the first one, will represent some intuitive phenotype, like resistance to fluconazole.

Moreover, we see many non-linearities in our data. For example, fitness in a double drug environment is not predicted by adding up fitness in the relevant single drug environments. Also, there are mutants that have high fitness when fluconazole is absent or abundant, but low fitness when mild concentrations are present. These types of nonlinearities can make the axes in PCA very difficult to interpret, plus these nonlinearities can be missed by PCA, thus we prefer other clustering methods.

We will adjust our revised manuscript to explain these reasons why we chose UMAP and GMM over PCA.

Also, we will include PCA in the supplement of our revised manuscript. Please find below PC1 vs PC2, with points colored according to the cluster assignment in figure 4 (i.e. using a gaussian mixture model). It appears the clusters are largely preserved.

**Author response image 1.**



*Second, the advantages of dimensional reduction are not clear. In the competition experiment, 11/12 conditions (all but the no drug, no DMSO conditions) can be mapped to only three dimensions: concentration of fluconazole, concentration of radicicol, and relative fitness. Each lineage would have its own fitness landscape as defined by the plane formed by relative fitness values in this space, which can then be examined and compared between lineages.*

We worry that the idea stems from apriori notions of what the important dimensions should be. It also seems like this would miss important nonlinearities such as our observation that low fluconazole behaves more like a novel selection pressure than a dialed down version of high fluconazole.

Also, we believe the reviewer meant “fitness profile” and not “fitness landscape”. A fitness landscape imagines a walk where every “step” is a mutation. Most lineages in barcoded evolution experiments possess only a single adaptive mutation. A single-step walk is not enough to build a landscape, though others are expanding barcoded evolution experiments beyond the first step (PMID34465770, PMID31723263), so maybe one day this will be possible.

*Third, the choice of 7 clusters as the cutoff for the multiple Gaussian model is not well explained. Based on Figure S6A, BIC starts leveling off at 6 clusters, not 7, and going to 8 clusters would provide the same reduction as going from 6 to 7. This choice also appears arbitrary in Figure S6B, where BIC levels off at 9 clusters when only highly abundant lineages are considered.*

We agree. We did not rely on the results of BIC alone to make final decisions about how many clusters to include. We thank the reviewer for pointing out this gap in our writing. We will adjust our revised manuscript to explain that we ultimately chose to describe 6 clusters that we were able to validate with follow-up experiments. In figures 5, 6, 7, and 8, we use external information to validate the clusters that we report in figure 4. And in lines 697 – 714, we explain that there are may be additional clusters beyond those we tease apart in this study.

*This directly contradicts the statement in the main text that clusters are robust to noise, as more a stringent inclusion threshold appears to increase and not decrease the optimal number of clusters. Additional criteria to BIC could have been used to help choose the optimal number of clusters or even if mixed Gaussian modeling is appropriate for this dataset.*

We are under the following impression: If our clustering method was overfitting, i.e. capturing noise, the optimal number of clusters should decrease when we eliminate noise. It increased. In other words, the observation that our clusters did not collapse (i.e. merge) when we removed noise suggests these clusters were not capturing noise.

More generally, our validation experiments, described below, provide additional evidence that our clusters capture meaningful differences between mutants (and not noise).

*(5) Large-scale barcode sequencing assays can often be noisy and are generally validated using growth curves or competition assays.*

Some types of bar-seq methods, in particular those that look at fold change across two time points, are noisier than others that look at how frequency changes across multiple timepoints (PMID30391162). Here, we use the less noisy method. We also reduce noise by using a stricter coverage threshold than previous work (e.g., PMID33263280), and by excluding batch effects by performing all experiments simultaneously (PMID37237236).

The main assay we use to measure fitness has been previously validated (PMID27594428). No subsequent study using this assay validates using the methods suggested by the reviewer (see PMID37861305, PMID33263280, PMID31611676, PMID29429618, PMID37192196, PMID34465770, PMID33493203).

More to the point, bar-seq has been used, without the reviewer’s suggested validation, to demonstrate that the way some mutant’s fitness changes across environments is different from other mutants (PMID33263280, PMID37861305, PMID31611676, PMID33493203, PMID34596043). This is the same thing that we use bar-seq to demonstrate.

For all of these reasons, we are hesitant to confirm bar-seq itself as a valid way to infer fitness. It seems this is already accepted as a standard in our field.



*Having these types of results would help support the accuracy of the main assay in the manuscript and thus better support the claims of the authors.*

We don't agree that fitness measurements obtained from this bar-seq assay generally require validation. But we do agree that it is important to validate whether the mutants in each of our 6 clusters indeed are different from one another in meaningful ways, in particular, in that they have different fitness tradeoffs. We have four figures (5 - 8) and 200 lines of text dedicated to validating whether our clusters capture reproducible and biologically meaningful differences between mutants. Happily, one of these figures (Fig 7) includes growth curves, which are exactly the type of validation experiment asked for by the reviewer.

Below, we walk through the different types of validation experiments that are present in our original manuscript, and additional validation experiments that we plan to include in the revised version. We are hopeful that these validation experiments are sufficient, or at the very least, that this list empowers reviewers to point out where more work is needed.

(1) Mutants from different clusters have different growth curves: In our original manuscript, we measured growth curves corresponding to a fitness tradeoff that we thought was surprising. Mutants in clusters 4 and 5 both have fitness advantages in single drug conditions. While mutants from cluster 4 also are advantageous in the double drug conditions, mutants from cluster 5 are not! We validated these different behaviors by studying growth curves for a mutant from each cluster (Figures 7 and S10).

(2) Mutants from different clusters have different evolutionary origins: In our original manuscript, we came up with a novel way to ask whether the clusters capture different types of adaptive mutants. We asked whether the mutants in each cluster originate from different evolution experiments. Indeed they often do (see pie charts in Figures 6, 7, 8). This method also provides evidence supporting each cluster's differing fitness tradeoffs.

For example, mutants in cluster 5 appear to have a tradeoff in a double drug condition (described above). They rarely originate from that evolution condition, unlike mutants in nearby cluster 4 (see Figure 7).

(3) Mutants from each cluster often fall into different genes: In our original manuscript, we sequenced many of these mutants and show that mutants in the same gene are often found in the same cluster. For example, all 3 IRA1 mutants are in cluster 6 (Fig 8), both GPB2 mutants are in cluster 4 (Figs 7 & 8), and 35/36 PDR mutants are in either cluster 2 or 3 (Figs 5 & 6).

(4) Mutants from each cluster have behaviors previously observed in the literature: In our original manuscript, we compared our sequencing results to the literature and found congruence. For example, PDR mutants are known to provide a fitness benefit in fluconazole and are found in clusters that have high fitness in fluconazole (lines 457 - 462). Previous work suggests that some mutations to PDR have different tradeoffs than others, which is what we see (lines 540 - 542). IRA1 mutants were previously observed to have high fitness in our "no drug" condition, and are found in the cluster that has the highest fitness in the "no drug" condition (lines 642 - 646). Previous work even confirms the unusual fitness tradeoff we observe where IRA1 and other cluster 6 mutants have low fitness only in low concentrations of fluconazole (lines 652 - 657).

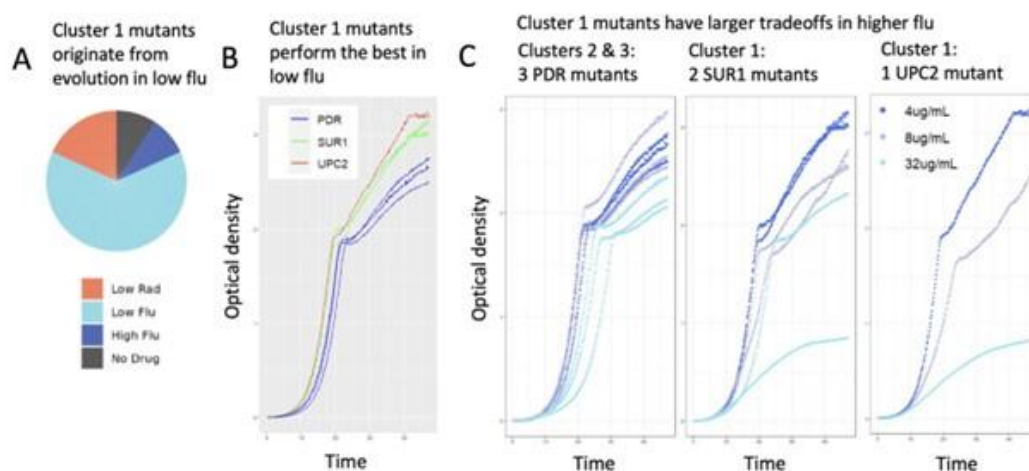
(5) Mutants largely remain in their clusters when we use alternate clustering methods: In our original manuscript, we performed various different reclustering and/or normalization approaches on our data (Fig 6, S5, S7, S8, S9). The clusters of mutants that we observe in figure 4 do not change substantially when we recluster the data. We will add PCA (see above) to these analyses in our revised manuscript.

(6) We will include additional data showing that mutants in different clusters have different evolutionary origins: Cluster 1 is defined by high fitness in low fluconazole that declines with increasing fluconazole (see Fig 4E and Fig 5C). In our revised manuscript, we will show that cluster 1 lineages were overwhelmingly sampled from evolutions conducted in our lowest concentration of fluconazole (see figure panel A below). No other cluster's evolutionary history shows this pattern (figures 6, 7, and 8).

(7) We will include additional data showing that mutants in different clusters have different growth curves: Cluster 1 lineages are unique in that their fitness advantage is specific to low flu and trades off in higher concentrations of fluconazole. We obtained growth curves for three cluster 1 mutants (2 SUR1 mutants and 1 UPC2 mutant). We compared them to growth curves for three PDR mutants (from clusters 2 and 3). Cluster 1 mutants appear to have the highest growth rates and reach the higher carrying capacity in low fluconazole (see red and green lines in Author response image 2 panel B below). But the cluster 1 mutants are negatively affected by higher concentrations of fluconazole, much more so than the mutants from clusters 2 and 3 (see Author response image 2 panel C below). This is consistent with the different fitness tradeoffs we observe for each cluster (figures 4 and 5). We will include a more detailed version of this analysis and the figures below in our revised manuscript.

### Author response image 2.

Validation experiments demonstrate that cluster 1 mutants have uniquely high fitness in only the lowest concentration of fluconazole. (A) The mutant lineages in cluster 1 were largely sampled from evolution experiments performed in low flu. This is not true of other clusters (see pie charts in main manuscript). (B) In low flu (4  $\mu\text{g/ml}$ ), Cluster 1 lineages (red/UPC2 and green/SUR1) grow faster and achieve higher density than lineages from clusters 2 and 3 (blue/PDR). This is consistent with barseq measurements demonstrating that cluster 1 mutants have the highest fitness in low flu. (C) Cluster 1 lineages are sensitive to increasing flu concentrations (SUR1 and UPC2 mutants, middle and rightmost graphs). This is apparent in that the gray (8  $\mu\text{g/ml}$  flu) and light blue (32  $\mu\text{g/ml}$  flu) growth curves rise more slowly and reach lower density than the dark blue curves (4  $\mu\text{g/ml}$  flu). But this is not the case for the PDR mutants from clusters 2 and 3 (leftmost graph). These observations are consistent with the bar-seq fitness data presented in the main manuscript (Fig 4E).



With all of these validation efforts combined, we are hopeful that the reviewer is now more convinced that our clusters capture groups of mutants with different fitness tradeoffs (as

opposed to noise). We want to conclude by saying that we are grateful to the reviewer for making us think deeply about areas where we can include additional validation efforts as well as areas where we can make our manuscript clearer.

**Reviewer #2 (Public Review):**

*Summary:*

*Schmidlin & Apodaca et al. aim to distinguish mutants that resist drugs via different mechanisms by examining fitness tradeoffs across hundreds of fluconazole-resistant yeast strains. They barcoded a collection of fluconazole-resistant isolates and evolved them in different environments with a view to having relevance for evolutionary theory, medicine, and genotype-phenotype mapping.*

*Strengths:*

*There are multiple strengths to this paper, the first of which is pointing out how much work has gone into it; the quality of the experiments (the thought process, the data, the figures) is excellent. Here, the authors seek to induce mutations in multiple environments, which is a really large-scale task. I particularly like the attention paid to isolates with are resistant to low concentrations of FLU. So often these are overlooked in favour of those conferring MIC values >64/128 etc. What was seen is different genotype and fitness profiles. I think there's a wealth of information here that will actually be of interest to more than just the fields mentioned (evolutionary medicine/theory).*

We are very grateful for this positive review. This was indeed a lot of work! We are happy that the reviewer noted what we feel is a unique strength of our manuscript: that we survey adaptive isolates across multiple environments, including low drug concentrations.

*Weaknesses:*

*Not picking up low fitness lineages - which the authors discuss and provide a rationale as to why. I can completely see how this has occurred during this research, and whilst it is a shame I do not think this takes away from the findings of this paper. Maybe in the next one!*

We thank the reviewer for these words of encouragement and will work towards catching more low fitness lineages in our next project.

*In the abstract the authors focus on 'tradeoffs' yet in the discussion they say the purpose of the study is to see how many different mechanisms of FLU resistance may exist (lines 679-680), followed up by "We distinguish mutants that likely act via different mechanisms by identifying those with different fitness tradeoffs across 12 environments". Whilst I do see their point, and this is entirely feasible, I would like a bit more explanation around this (perhaps in the intro) to help lay-readers make this jump. The remainder of my comments on 'weaknesses' are relatively fixable, I think:*

We think that phrasing the “jump” as a question might help lay readers get from point A to point B. So, in the introduction of our revised manuscript, we will add a paragraph roughly similar to this one: “If two groups of drug-resistant mutants have different fitness tradeoffs, does it mean that they provide resistance through different underlying mechanisms? Alternatively, it could mean that both provide drug resistance via the same mechanism, but some mutations come with a cost that others don’t pay. However, another way to phrase this alternative is to say that both groups of mutants affect fitness through different suites of mechanisms that are only partially overlapping. And so, by identifying groups of mutants with different fitness tradeoffs, we argue that we will be uncovering sets of mutations that

impact fitness through different underlying mechanisms. The ability to do so would be useful for genotype-phenotype mapping endeavors.”

*In the introduction I struggle to see how this body of research fits in with the current literature, as the literature cited is a hodge-podge of bacterial and fungal evolution studies, which are very different! So example, the authors state "previous work suggests that mutants with different fitness tradeoffs may affect fitness through different molecular mechanisms" (lines 129-131) and then cite three papers, only one of which is a fungal research output. However, the next sentence focuses solely on literature from fungal research. Citing bacterial work as a foundation is fine, but as you're using yeast for this I think tailoring the introduction more to what is and isn't known in fungi would be more appropriate. It would also be great to then circle back around and mention monotherapy vs combination drug therapy for fungal infections as a rationale for this study. The study seems to be focused on FLU-resistant mutants, which is the first-line drug of choice, but many (yeast) infections have acquired resistance to this and combination therapy is the norm.*

In our revised manuscript, we will carefully review all citations. The issue may stem from our attempt to reach two different groups of scientists. We ourselves are broadly interested in the structure of the genotype-phenotype-fitness map (PMID33263280, PMID32804946). Though the 3 papers the reviewer mentions on lines 132 - 133 all pertain to yeast, we cite them because they are studies about the complexity of this map. Their conclusions, in theory, should apply broadly, beyond yeast. Similarly, the reason we cite papers from yeast, as well as bacteria and cancer, is that we believe general conclusions about the genotype-phenotype-fitness map should apply broadly. For example, the sentence the reviewer highlights, “previous work suggests that mutants with different fitness tradeoffs may affect fitness through different molecular mechanisms” is a general observation about the way genotype maps to fitness. So we cited papers from across the tree of life to support this sentence.

On the other hand, because we study drug resistant mutations, we also hope that our work is of use to scientists studying the evolution of resistance. We agree with the reviewer that in this regard, some of our findings may be especially pertinent to the evolution of resistance to antifungal drugs. We will consider this when reviewing the citations in our revised manuscript and add some text to clarify these points.

*Methods: Line 769 - which yeast? I haven't even seen mention of which species is being used in this study; different yeast employ different mechanisms of adaptation for resistance, so could greatly impact the results seen. This could help with some background context if the species is mentioned (although I assume *S. cerevisiae*).*

In the revised manuscript, we will make clear that we study *S. cerevisiae*.

*In which case, should aneuploidy be considered as a mechanism? This is mentioned briefly on line 556, but with all the sequencing data acquired this could be checked quickly?*

We like this idea and we are working on it, but it is not straightforward. The reviewer is correct in that we can use the sequencing data that we already have. But calling aneuploidy with certainty is tough because its signal can be masked by noise. In other words, some regions of the genome may be sequenced more than others by chance. Given this is not straightforward, at least not for us, this analysis will likely have to wait for a subsequent paper.

*I think the authors could be bolder and try and link this to other (pathogenic) yeasts.  
What are the implications of this work on say, Candida infections?*

Perhaps because our background lies in general study of the genotype-phenotype map, we did not want to make bold assertions about how our work might apply to pathogenic yeasts. But we see how this could be helpful and will add some discussion points about this. Specifically, we will discuss which of the genes and mutants we observe are also found in *Candida*. We will also investigate whether our observation that low fluconazole represents a seemingly unique challenge, not just a milder version of high fluconazole, has any corollary in the *Candida* literature.